



Aplicações de Mapas de Karnaugh

Circuitos Digitais I
Prof. Fernando Passold

Resumo de Mapas de Karnaugh:

- Agrupamentos de **2** células:

AB \ C	00	01
00		
01	1	
11	1	
10		

$$x = B\bar{C}$$

AB \ C	00	01
00		
01	1	1
11		
10		

$$x = \bar{A}B$$

AB \ C	00	01
00	1	
01		
11		
10	1	

$$x = \bar{B}\bar{C}$$

AB \ CD	00	01	11	10
00			1	1
01				
11				
10	1			1

$$x = \bar{A}\bar{B}C + A\bar{B}\bar{D}$$

Resumo de Mapas de Karnaugh:

- Agrupamentos de 4 células:

AB \ C	00	01
00		1
01		1
11		1
10		1

$$x = C$$

AB \ CD	00	01	11	10
00				
01				
11	1	1	1	1
10				

$$x = AB$$

AB \ CD	00	01	11	10
00				
01		1	1	
11		1	1	
10				

$$x = BD$$

AB \ CD	00	01	11	10
00				
01				
11	1			1
10	1			1

$$x = A\bar{D}$$

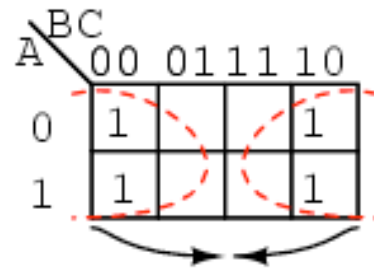
AB \ CD	00	01	11	10
00	1			1
01				
11				
10	1			1

$$x = \bar{B}\bar{D}$$

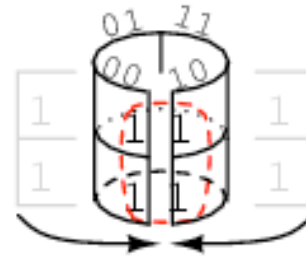
Resumo de Mapas de Karnaugh:

- Agrupamentos de 4 células:

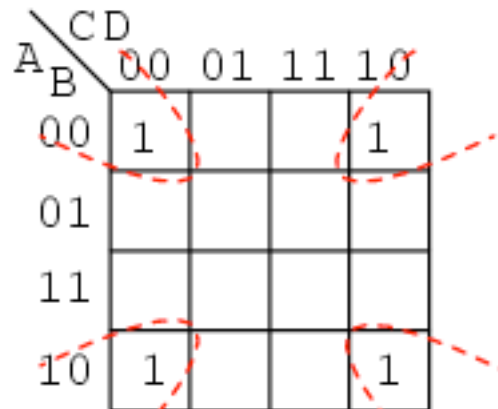
$$\text{Out} = \overline{A}\overline{B}\overline{C} + A\overline{B}\overline{C} + \overline{A}B\overline{C} + AB\overline{C}$$



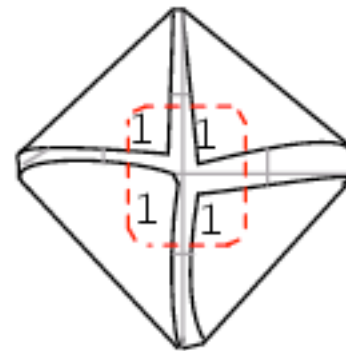
$$\text{Out} = \overline{C}$$



$$\text{Out} = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + A\overline{B}\overline{C}\overline{D} + A\overline{B}C\overline{D}$$



$$\text{Out} = \overline{B}\overline{D}$$



Resumo de Mapas de Karnaugh:

- Agrupamentos de **8** células:

AB \ CD	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10				

$$x = B$$

AB \ CD	00	01	11	10
00	1	1		
01	1	1		
11	1	1		
10	1	1		

$$x = \bar{C}$$

AB \ CD	00	01	11	10
00	1	1	1	1
01				
11				
10	1	1	1	1

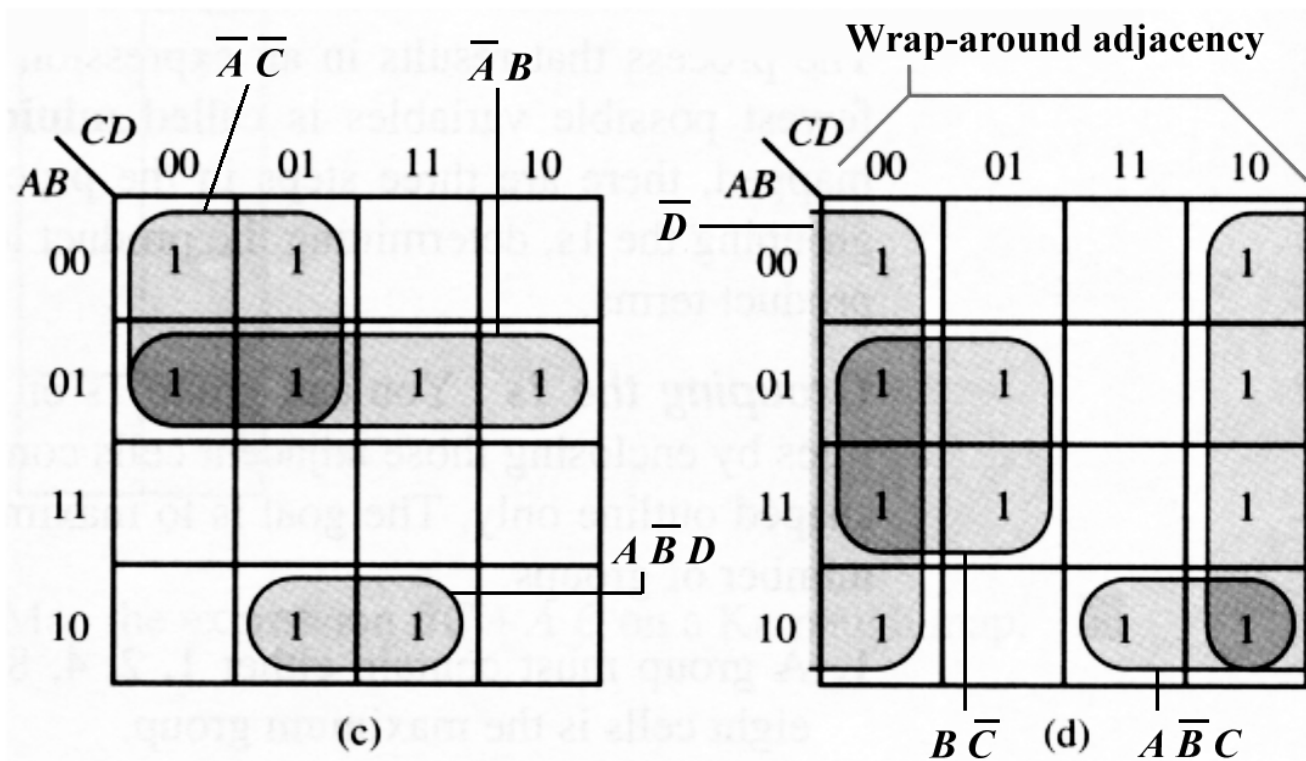
$$x = \bar{B}$$

AB \ CD	00	01	11	10
00	1			1
01	1			1
11	1			1
10	1			1

$$x = \bar{D}$$

Mapas de Karnaugh:

- Exemplos de Simplificações possíveis:



Mapas de Karnaugh:

- Exemplos de Simplificações possíveis:

AB\CD	00	01	11	10
00				1
01		1	1	
11		1	1	
10				

$$X = \underbrace{\bar{A}\bar{B}\bar{C}\bar{D}}_{\text{loop 4}} + \underbrace{ACD}_{\text{loop 11, 15}} + \underbrace{BD}_{\text{loop 6, 7, 10, 11}}$$

(a)

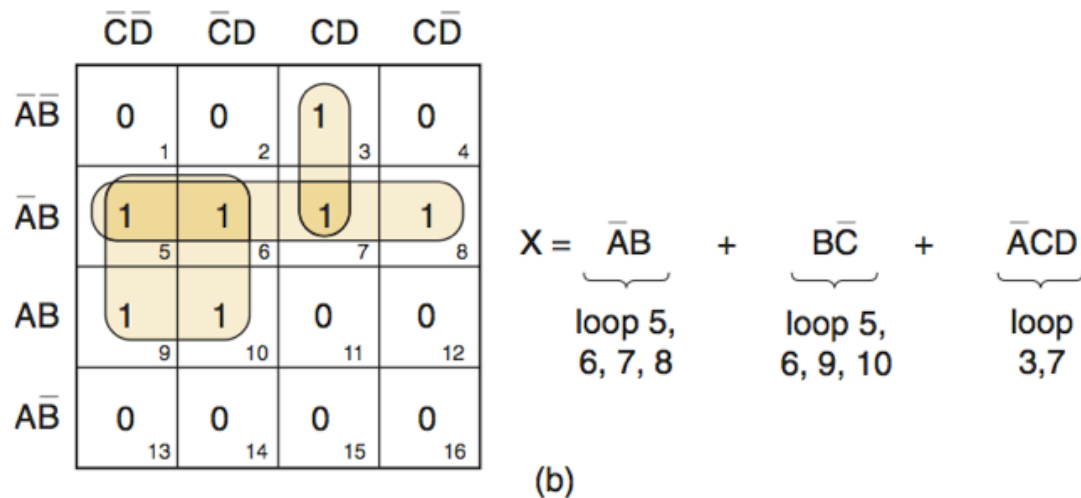
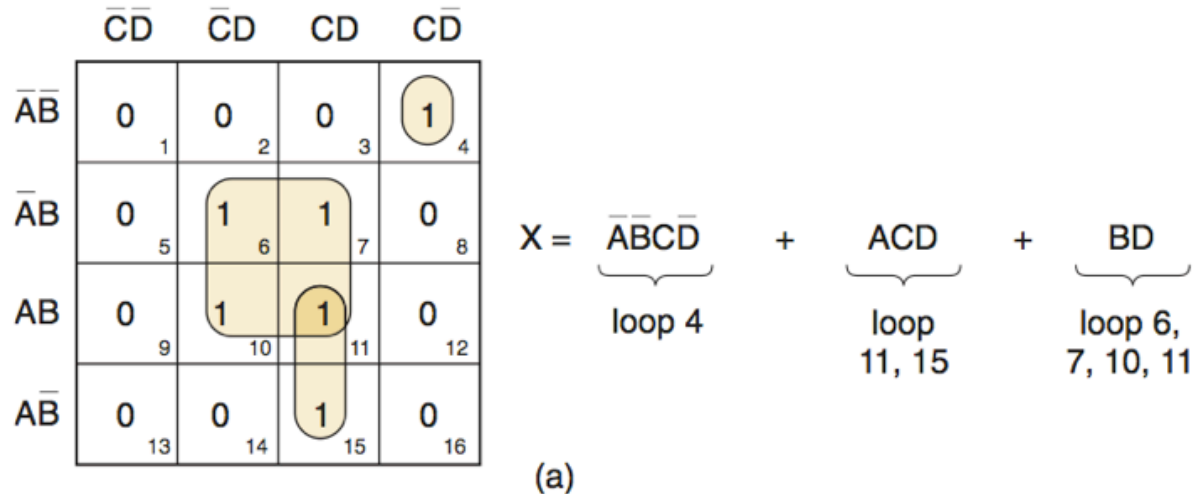
AB\CD	00	01	11	10
00			1	
01	1	1	1	1
11	1	1		
10				

$$X = \underbrace{\bar{A}\bar{B}}_{\text{loop 5, 6, 7, 8}} + \underbrace{B\bar{C}}_{\text{loop 5, 6, 9, 10}} + \underbrace{\bar{A}CD}_{\text{loop 3, 7}}$$

(b)

Mapas de Karnaugh:

- Exemplos de Simplificações possíveis:



Mapas de Karnaugh:

- Exemplos de Simplificações possíveis:

AB\CD	00	01	11	10
00		1		
01		1	1	1
11	1	1	1	
10			1	

Mapas de Karnaugh:

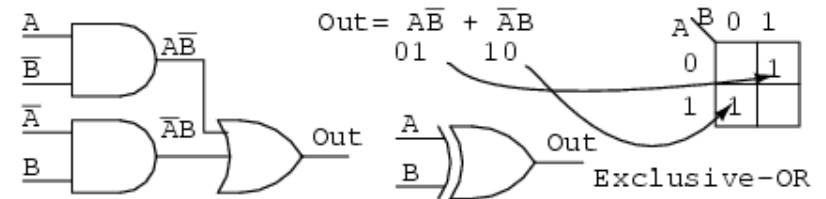
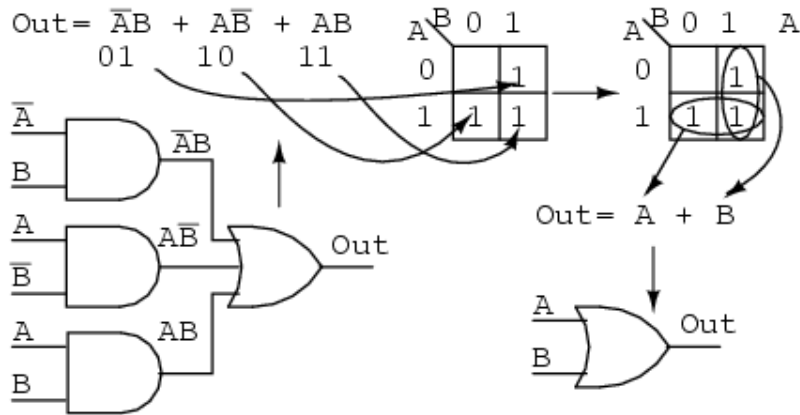
- Exemplos de Simplificações possíveis:

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0 1	1 2	0 3	0 4
$\bar{A}B$	0 5	1 6	1 7	1 8
AB	1 9	1 10	1 11	0 12
$A\bar{B}$	0 13	0 14	1 15	0 16

$$X = \underbrace{ABC}_{9, 10} + \underbrace{\bar{A}\bar{C}D}_{2, 6} + \underbrace{\bar{A}BC}_{7, 8} + \underbrace{ACD}_{11, 15}$$

(c)

Exemplos de Simplificações...



$$\text{Out} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + ABCD$$

A \ B	00	01	11	10
00	1			1
01				
11			1	
10	1			

$$\text{Out} = \bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{D} + ABCD$$

$$\begin{aligned} \text{Out} = & \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD \\ & + AB\bar{C}\bar{D} + ABCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} \end{aligned}$$

A \ B	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10	1			1

$$\text{Out} = \bar{B}\bar{C}\bar{D} + \bar{A}\bar{C}\bar{D} + BCD + AC\bar{D}$$

$$\text{Out} = \bar{A}\bar{B}\bar{C} + \bar{A}BD + ABC + A\bar{B}\bar{D}$$

A \ B	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10	1			1

Exemplos de Simplificações...

$$\text{Out} = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + ABCD$$

A \ B \ CD	00	01	11	10
00	1			1
01				
11			1	
10	1			

$$\text{Out} = \overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{D} + ABCD$$

$$\text{Out} = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}BCD + ABCD + \overline{A}BC\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D}$$

A \ B \ CD	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10	1			1

$$\text{Out} = \overline{B}\overline{C}\overline{D} + \overline{A}\overline{C}\overline{D} + BCD + AC\overline{D}$$

$$\text{Out} = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}D + ABC + A\overline{B}\overline{D}$$

A \ B \ CD	00	01	11	10
00	1	1		
01		1	1	
11			1	1
10	1			1

$$\begin{aligned} \text{Out} = & \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}\overline{B}CD \\ & + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BCD \\ & + AB\overline{C}\overline{D} + AB\overline{C}D + ABCD \end{aligned}$$

A \ B \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10				

A \ B \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10				

A \ B \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10				

$$\text{Out} = \overline{A}\overline{C} + \overline{A}D + B\overline{C} + BD$$

$$\begin{aligned} \text{Out} = & \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}BC\overline{D} + \overline{A}BCD \\ & + AB\overline{C}\overline{D} + AB\overline{C}D + ABCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}C\overline{D} \end{aligned}$$

A \ B \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10	1			1

$$\text{Out} = \overline{C}\overline{D} + CD + \overline{A}B\overline{C}$$

A \ B \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10	1			1

$$\text{Out} = \overline{C}\overline{D} + CD + ABD$$

Mapas de Karnaugh:

- Exemplos de Simplificações possíveis:

$$\begin{aligned} \text{Out} = & \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} \\ & + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + \overline{A}BC\overline{D} \\ & + AB\overline{C}\overline{D} + AB\overline{C}D + ABC\overline{D} \end{aligned}$$

		CD			
		00	01	11	10
A/B	00	1	1	1	
	01	1	1	1	
	11	1	1	1	
	10				

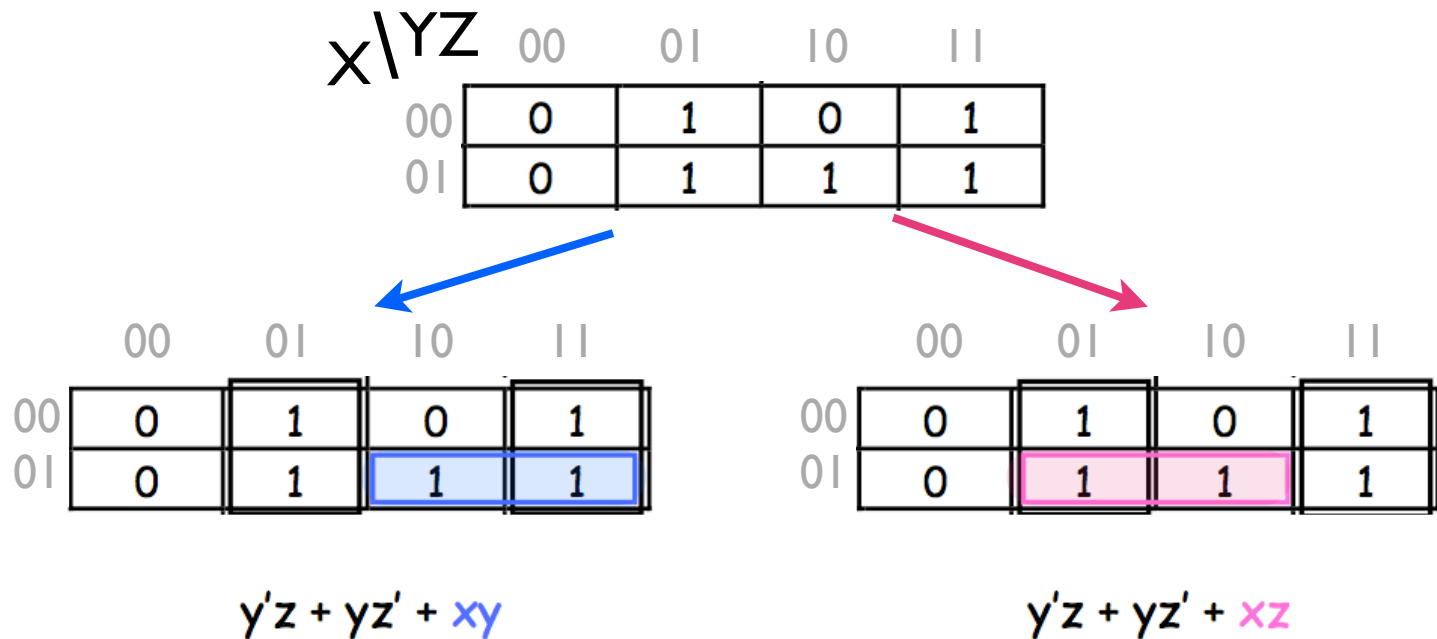
		CD			
		00	01	11	10
A/B	00	1	1	1	
	01	1	1	1	
	11	1	1	1	
	10				

		CD			
		00	01	11	10
A/B	00	1	1	1	
	01	1	1	1	
	11	1	1	1	
	10				

$$\text{Out} = \overline{A}\overline{C} + \overline{A}D + B\overline{C} + BD$$

Mapas de Karnaugh:

- Exemplo de 2 soluções igualmente possíveis:



Mapas de Karnaugh:

- Exemplo de 2 soluções igualmente possíveis:

$$\begin{aligned} \text{out} = & \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD \\ & + ABCD + ABC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} \end{aligned}$$

		CD			
		00	01	11	10
A/B	00	1	1		
	01		1	1	
	11			1	1
	10	1			1

$$\text{out} = \bar{B}\bar{C}\bar{D} + \bar{A}\bar{C}D + BCD + AC\bar{D}$$

$$\text{out} = \bar{A}\bar{B}\bar{C} + \bar{A}BD + ABC + A\bar{B}\bar{D}$$

		CD			
		00	01	11	10
A/B	00	1	1		
	01		1	1	
	11			1	1
	10	1			1

Mapas de Karnaugh:

- Exemplo de 2 soluções igualmente possíveis:

$$\text{Out} = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}BC\overline{D} + AB\overline{C}\overline{D} + AB\overline{C}D + ABCD + A\overline{B}\overline{C}\overline{D} + A\overline{B}CD$$

		CD			
		00	01	11	10
AB	00	1		1	
	01	1		1	
	11	1	1	1	
	10	1		1	

$$\text{Out} = \overline{C}\overline{D} + CD + AB\overline{C}$$

		CD			
		00	01	11	10
AB	00	1		1	
	01	1		1	
	11	1	1	1	
	10	1		1	

$$\text{Out} = \overline{C}\overline{D} + CD + ABD$$

Mapas de Karnaugh:

- Exemplo de 2 soluções igualmente muito boas:

AB \ CD	00	01	11	10
00		1		
01		1	1	1
11				1
10	1	1		1

$$X = \bar{A}\bar{C}D + \bar{A}BC + A\bar{B}\bar{C} + ACD$$

(a)

AB \ CD	00	01	11	10
00		1		
01		1	1	1
11				1
10	1	1		1

$$X = \bar{A}BD + BCD + \bar{B}\bar{C}D + A\bar{B}\bar{D}$$

(b)

Mapas de Karnaugh:

- Exemplo de 2 soluções igualmente muito boas:

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	0	0
$\bar{A}B$	0	1	1	1
AB	0	0	0	1
$A\bar{B}$	1	1	0	1

$$X = \bar{A}\bar{C}D + \bar{A}BC + A\bar{B}\bar{C} + ACD$$

(a)

Circuito (a):

1 x OR(4)+4xAND(3)+4xNOT

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	0	0
$\bar{A}B$	0	1	1	1
AB	0	0	0	1
$A\bar{B}$	1	1	0	1

$$X = \bar{A}BD + BCD + \bar{B}\bar{C}D + A\bar{B}\bar{D}$$

(b)

Circuito (b):

1 x OR(4)+4xAND(3)+4xNOT

Problema

- Projete o circuito lógico que possui 4 entradas (A, B, C e D) e que seja capaz de detectar as situações nas quais apenas 2 variáveis de entrada estão ativadas.
Obs: D=MSB, A=LSB.

Ref	DCBA	Z
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

$$Z = \bar{D}\bar{C}BA + (B \oplus A)(D \oplus C) + DC\bar{B}\bar{A}$$

Problema

- Projete o circuito lógico que possui 4 entradas (A, B, C e D) e que seja capaz de detectar as situações nas quais apenas 2 variáveis de entrada estão ativadas.
Obs: D=MSB, A=LSB.

Ref	DCBA	Z
0	0000	
1	0001	
2	0010	
3	0011	1
4	0100	
5	0101	1
6	0110	1
7	0111	
8	1000	
9	1001	1
10	1010	1
11	1011	
12	1100	1
13	1101	
14	1110	
15	1111	

$$= \bar{D}\bar{C}BA$$

$$= \bar{D}C\bar{B}A$$

$$= \bar{D}CBA\bar{A}$$

$$= D\bar{C}\bar{B}A$$

$$= D\bar{C}B\bar{A}$$

$$= DC\bar{B}\bar{A}$$

Z: dc\BA	00	01	11	10
00	0	1	1 ³	2
01	4	1 ⁵	7	1 ⁶
11	1 ¹²	13	15	14
10	8	1 ⁹	11	1 ¹⁰



$$Z = \bar{D}\bar{C}BA + (B \oplus A)(D \oplus C) + DC\bar{B}\bar{A}$$

Problema

- Projete o circuito lógico que possui 4 entradas (A, B, C e D) e que seja capaz de detectar as situações nas quais apenas 2 variáveis de entrada estão ativadas.
Obs: D=MSB, A=LSB.

Ref	DCBA	Z
0	0000	
1	0001	
2	0010	
3	0011	1
4	0100	
5	0101	1
6	0110	1
7	0111	
8	1000	
9	1001	1
10	1010	1
11	1011	
12	1100	1
13	1101	
14	1110	
15	1111	

$$= \bar{D}\bar{C}BA$$

$$= \bar{D}C\bar{B}A$$

$$= \bar{D}CBA\bar{A}$$

$$= D\bar{C}\bar{B}A$$

$$= D\bar{C}BA\bar{A}$$

$$= DC\bar{B}\bar{A}$$

Z: DC\BA	00	01	11	10
00	0	1	1 ³	2
01	4	1 ⁵	7	1 ⁶
11	1 ¹²	13	15	14
10	8	1 ⁹	11	1 ¹⁰

$$Z = \underbrace{\bar{D}\bar{C}BA}_{m_3} + \underbrace{\bar{D}C\bar{B}A}_{m_5} + \underbrace{\bar{D}CBA\bar{A}}_{m_6} + \underbrace{D\bar{C}\bar{B}A}_{m_9} + \underbrace{D\bar{C}BA\bar{A}}_{m_{10}} + \underbrace{DC\bar{B}\bar{A}}_{m_{12}}$$

$$Z = \bar{D}\bar{C}BA + \bar{D}C(\bar{B}A + B\bar{A}) + D\bar{C}(\bar{B}A + B\bar{A}) + DC\bar{B}\bar{A}$$

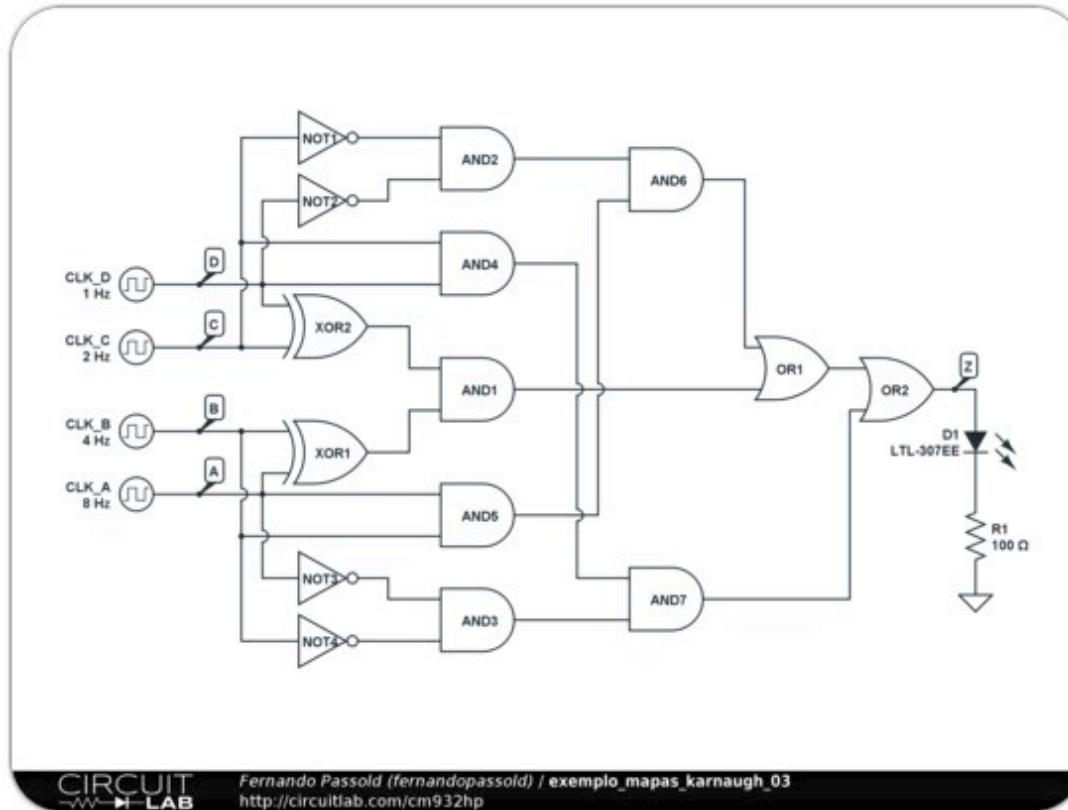
$$Z = \bar{D}\bar{C}BA + \bar{D}C(B \oplus A) + D\bar{C}(B \oplus A) + DC\bar{B}\bar{A}$$

$$Z = \bar{D}\bar{C}BA + (B \oplus A)(\bar{D}C + D\bar{C}) + DC\bar{B}\bar{A}$$

$$Z = \bar{D}\bar{C}BA + (B \oplus A)(D \oplus C) + DC\bar{B}\bar{A}$$

Problema

- Projete o circuito lógico que possui 4 entradas (A, B, C e D) e que seja capaz de detectar as situações nas quais apenas 2 variáveis de entrada estão ativadas. Obs: D=MSB. A=LSB.



- Solução --> Circuito: disponível em: https://www.circuitlab.com/circuit/m932hp/exemplo_mapas_karnaugh_03/ (em 13/set/2013)
- Uso do CircuitLab (web-based electronics design tool): www.circuitlab.com

DC

DC Sweep

Time Domain

Start Time: 0 s

Stop Time: 1.2 s

Time Step: 1 s

Skip Initial: No

Sweep Parameter: ☐

Outputs:

$V(D)/5+1.25$

$V(C)/5+2.5$

$V(B)/5+3.75$

$V(A)/5+5$

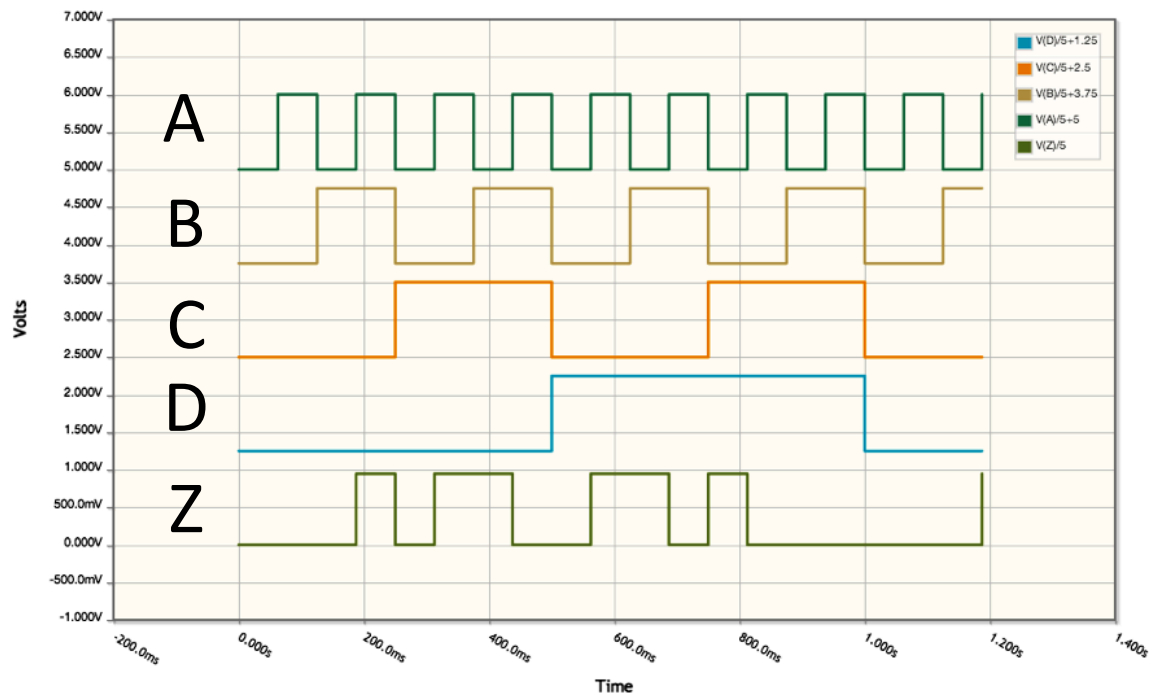
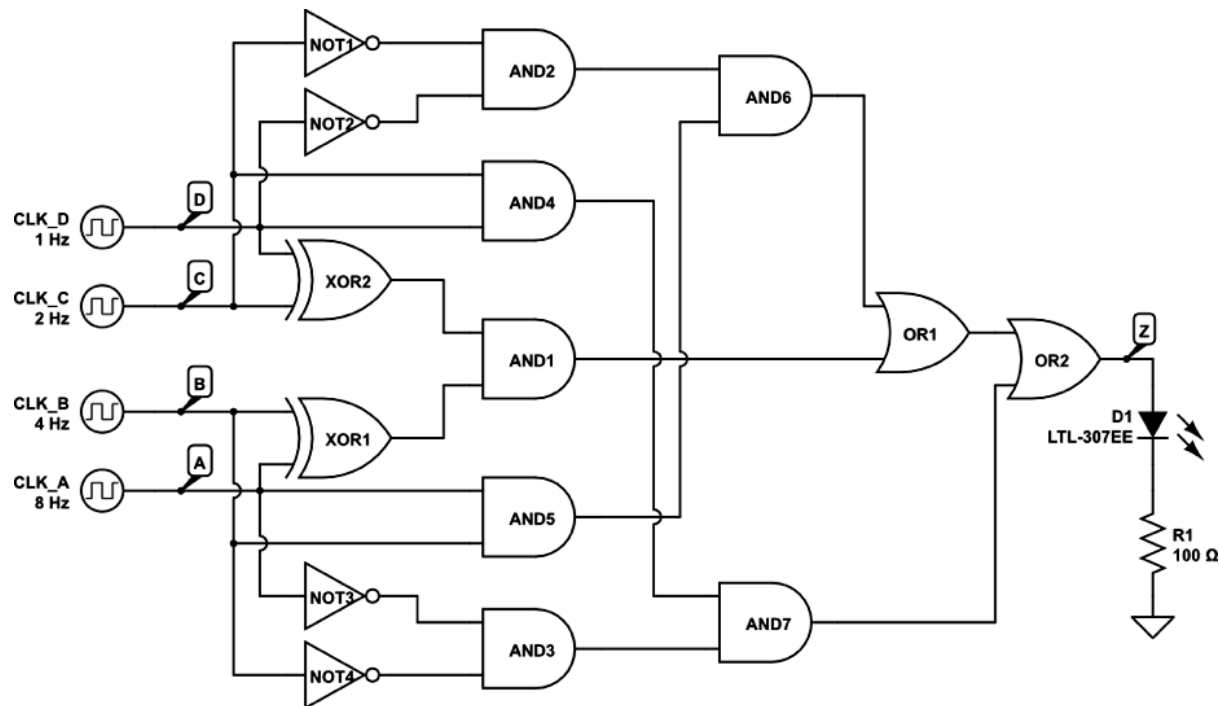
$V(Z)/5$

+ Add Expression

Advanced Graphing...

Run Time-Domain Simulation

Frequency Domain



Mapas K para 5 variáveis

2 opções:

Repare na seq. dos códigos

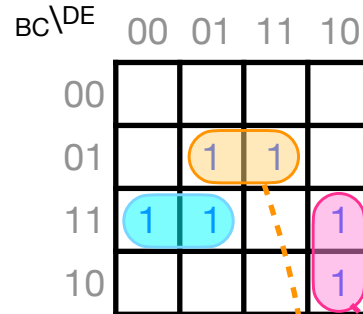
Repare:
A=bit mais significativo!

Tabela Verdade

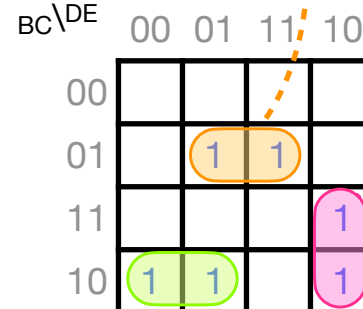
Ref	ABCDE	Y
0	00000	
1	1	
:	:	
15	01111	
16	10000	
17	10001	
:	:	
30	11110	
31	11111	

Método “camadas” (overlay)

A = 0

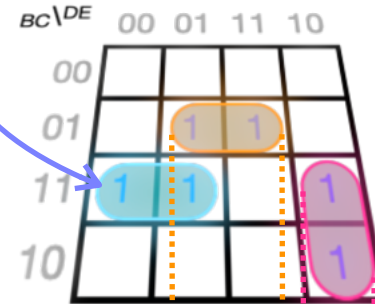


A = 1



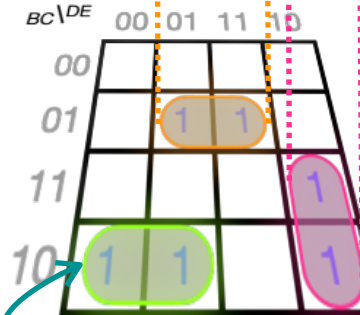
$$= \bar{A} \cdot B \cdot C \cdot \bar{D}$$

A = 0



$$\bar{B} \cdot C \cdot E =$$

A = 1



$$= B \cdot D \cdot \bar{E}$$

$$= A \cdot B \cdot \bar{C} \cdot \bar{D}$$

Mapas K para 5 variáveis

2 opções:

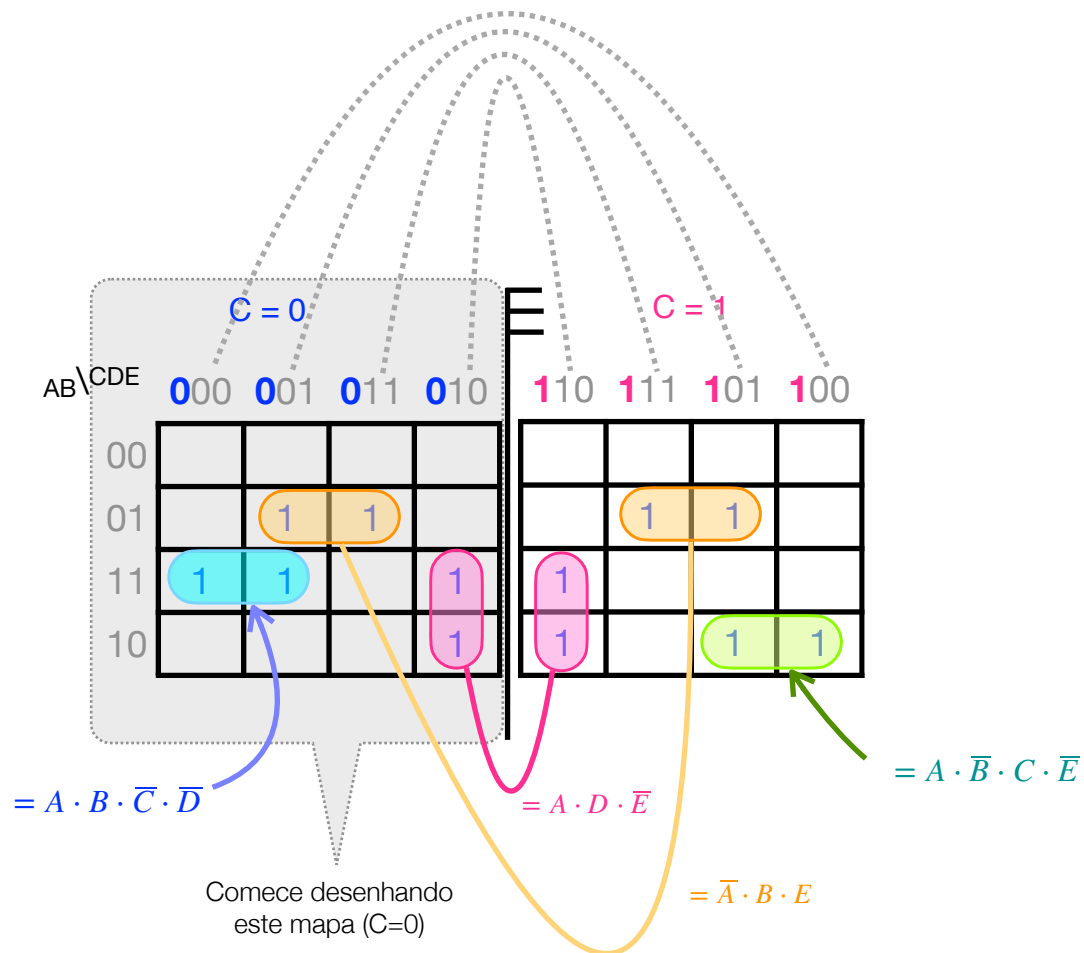
Repare na seq. dos códigos

Método “espelho” (mirror)

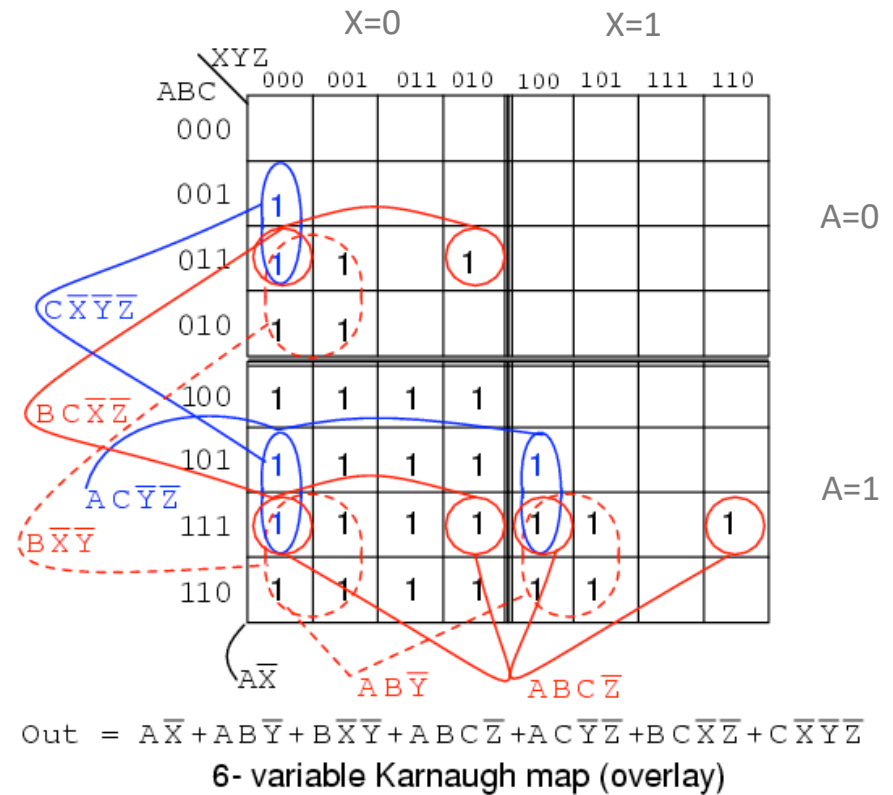
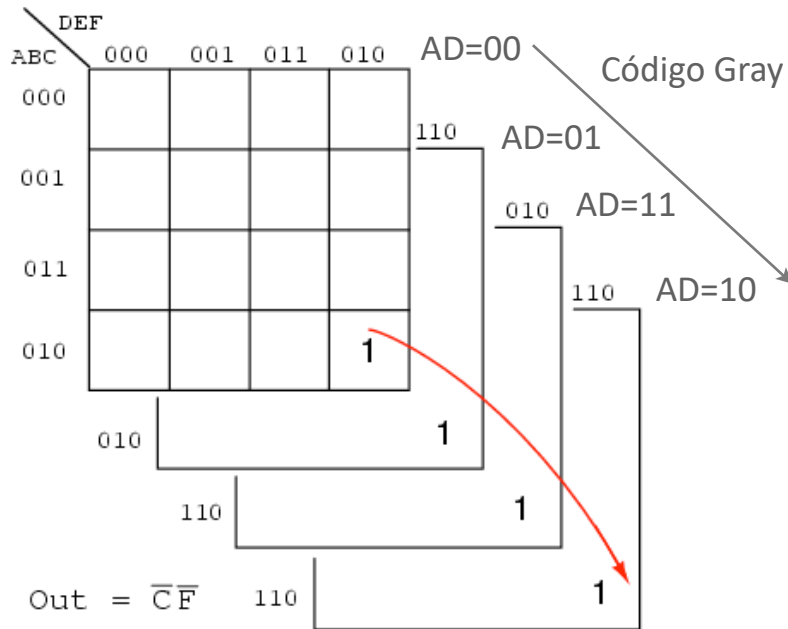
Repare:
A=bit mais significativo!

Tabela Verdade

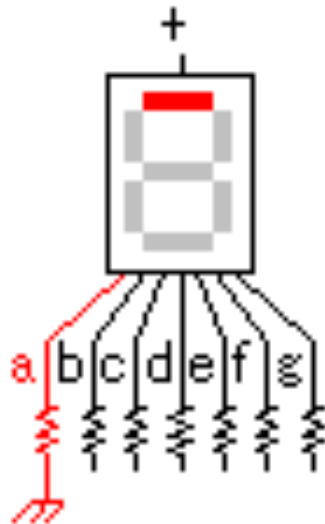
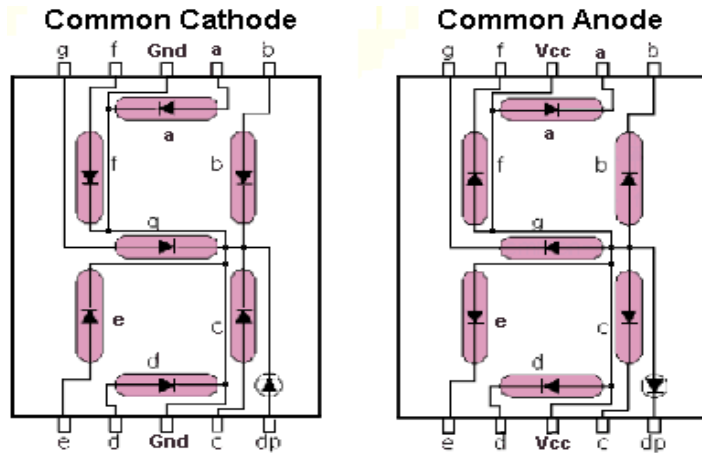
Ref	ABCDE	Y
0	00000	
1	1	
:	:	
15	01111	
16	10000	
17	10001	
:	:	
30	11110	
31	11111	



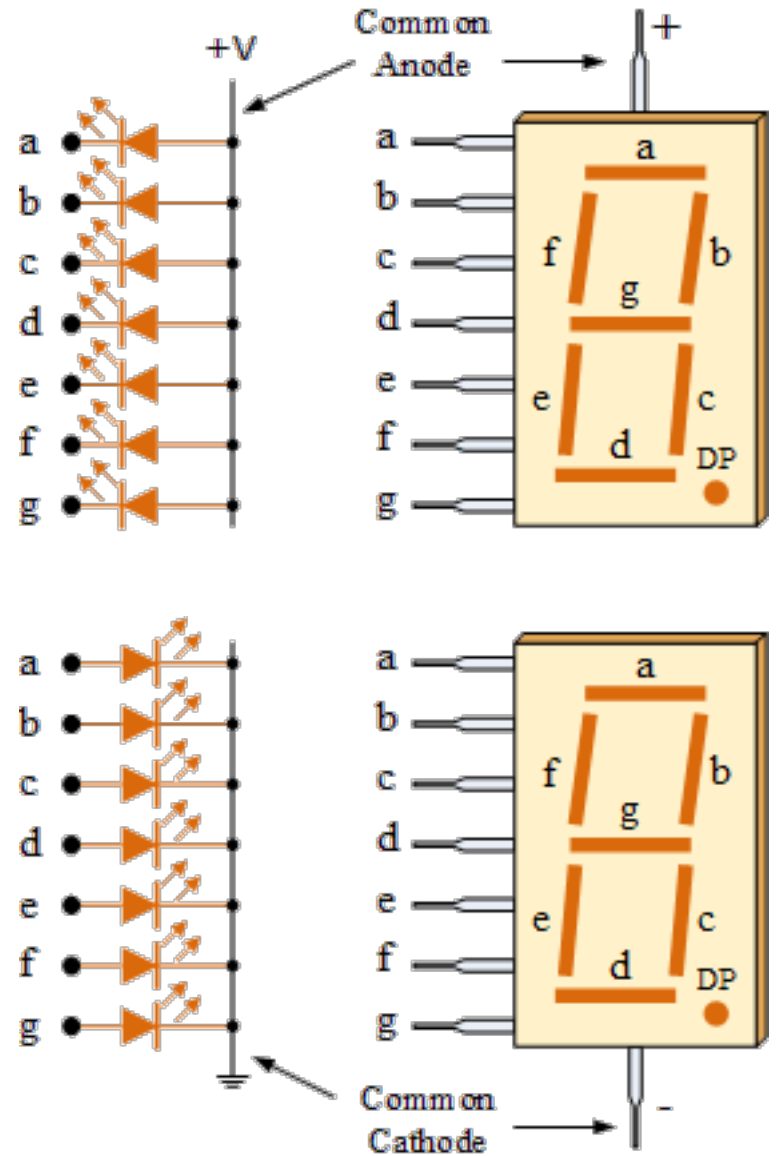
Mapas K para 6 variáveis



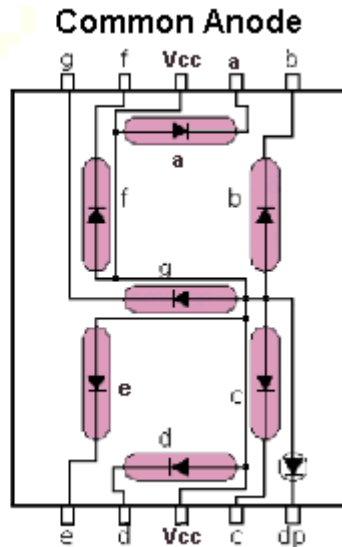
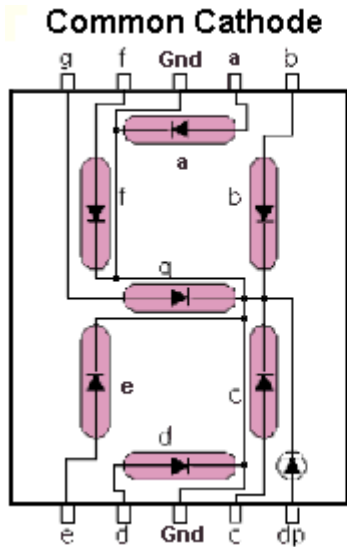
DEC Display 7 Segmentos



Ânodo comum



Display 7 Segmentos / BCD

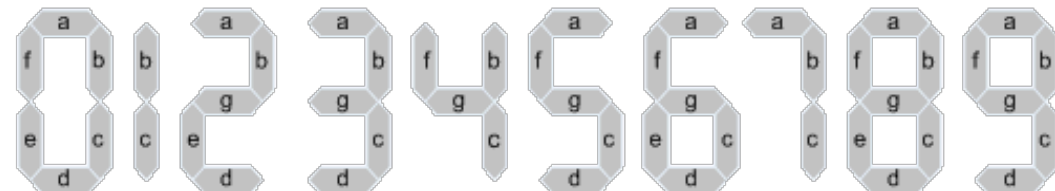


DECIMAL	DCBA
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

www.electronicsarea.com

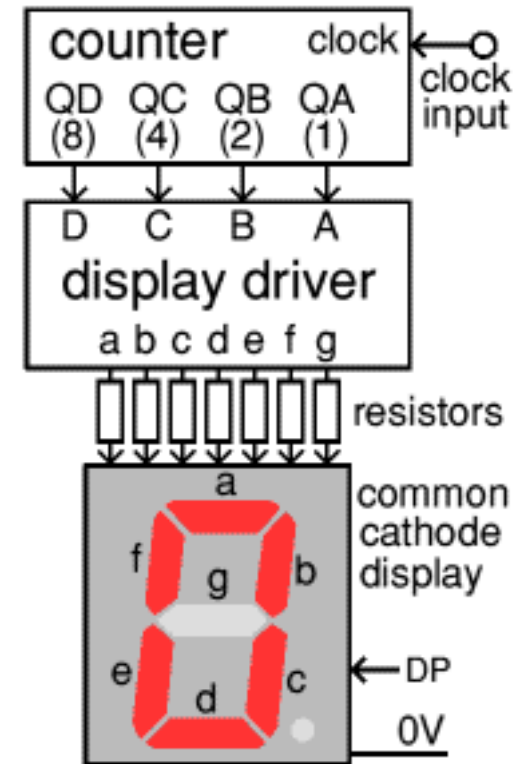
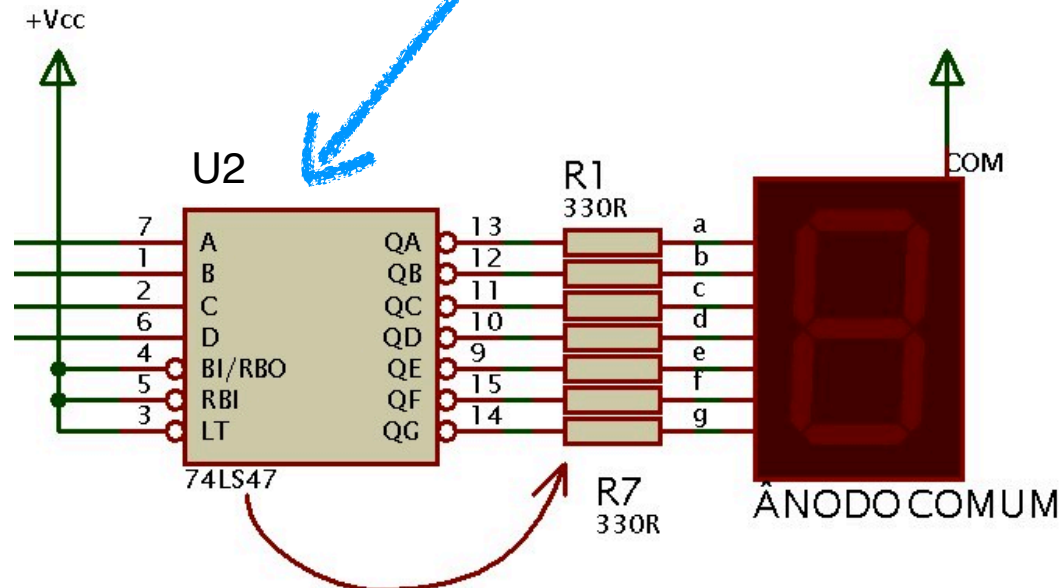
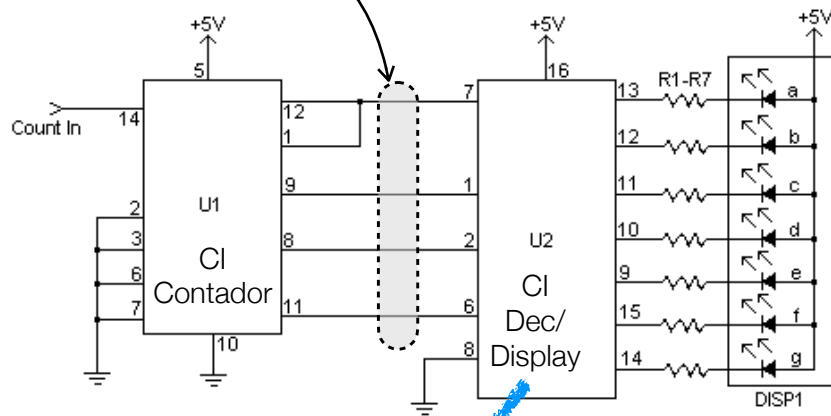
Digit Shown	Illuminated Segment (1 = illumination)						
	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	0	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

AbCdEfGhI jKlñ
 nOpq,rStUv'wXyZ
 0 1234567890



DEC Display 7

Gera seq. Binária (4-bits)



DEC Display



MOTOROLA

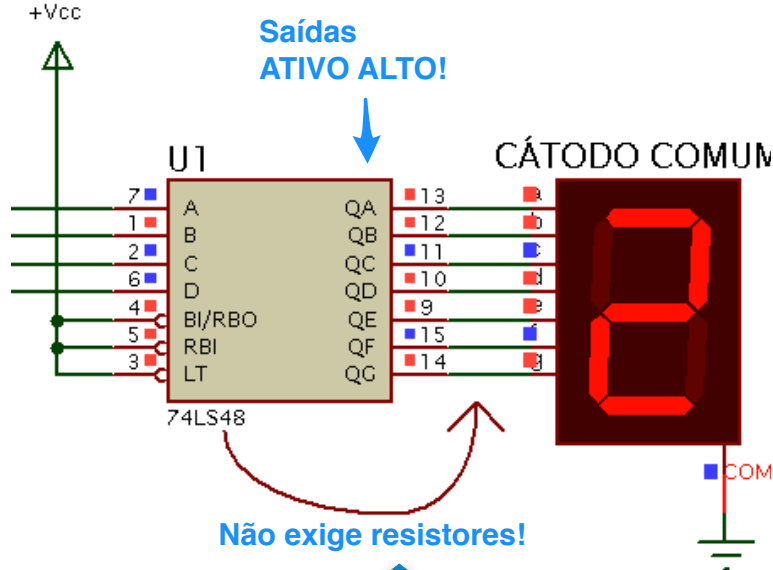
BCD TO 7-SEGMENT DECODER

The SN54/74LS48 is a BCD to 7-Segment Decoder consisting of NAND gates, input buffers and seven AND-OR-INVERT gates. Seven NAND gates and one driver are connected in pairs to make BCD data and its complement available to the seven decoding AND-OR-INVERT gates. The remaining NAND gate and three input buffers provide lamp test, blanking input/ripple-blanking input for the LS48.

The circuit accepts 4-bit binary-coded-decimal (BCD) and, depending on the state of the auxiliary inputs, decodes this data to drive other components. The relative positive logic output levels, as well as conditions required at the auxiliary inputs, are shown in the truth tables.

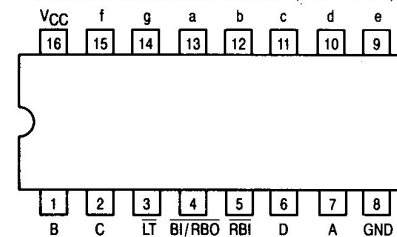
The LS48 circuit incorporates automatic leading and/or trailing edge zero-blanking control (RBI and RBO). Lamp Test (LT) may be activated any time when the BI/RBO node is HIGH. Both devices contain an overriding blanking input (BI) which can be used to control the lamp intensity by varying the frequency and duty cycle of the BI input signal or to inhibit the outputs.

- Lamp Intensity Modulation Capability (BI/RBO)
- Internal Pull-Ups Eliminate Need for External Resistors
- Input Clamp Diodes Eliminate High-Speed Termination Effects

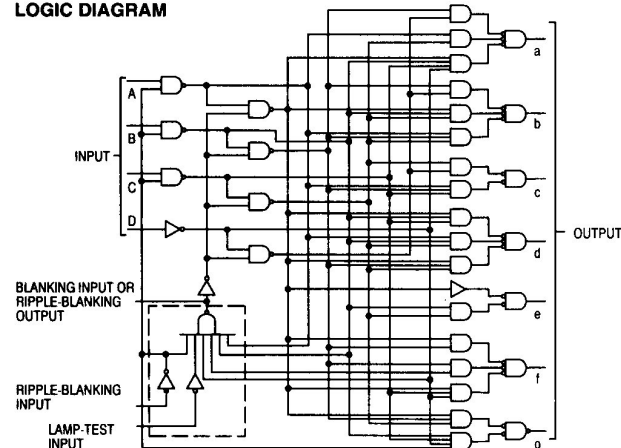


Obs.: mas apenas o CI 74LS48!

CONNECTION DIAGRAM DIP (TOP VIEW)



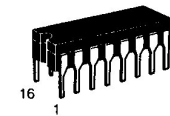
LOGIC DIAGRAM



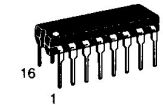
SN54/74LS48

BCD TO 7-SEGMENT DECODER

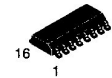
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

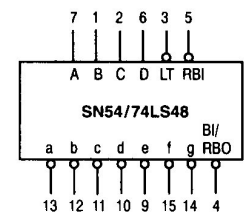


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

SN54LSXXJ Ceramic
SN74LSXXN Plastic
SN74LSXXD SOIC

LOGIC SYMBOL



VCC = PIN 16
GND = PIN 8

DEC Display 7

BCD TO 7-SEGMENT DECODER/DRIVER

The SN54/74LS47 are Low Power Schottky BCD to 7-Segment Decoder/Drivers consisting of NAND gates, input buffers and seven AND-OR-IN-

SN54/74LS47

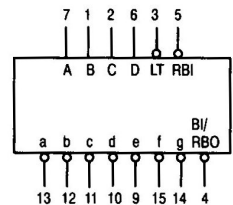
**BCD TO 7-SEGMENT
DECODER/DRIVER**
LOW POWER SCHOTTKY



ORDERING INFORMATION

SN54LSXXJ	Ceramic
SN74LSXXN	Plastic
SN74LSXXD	SOIC

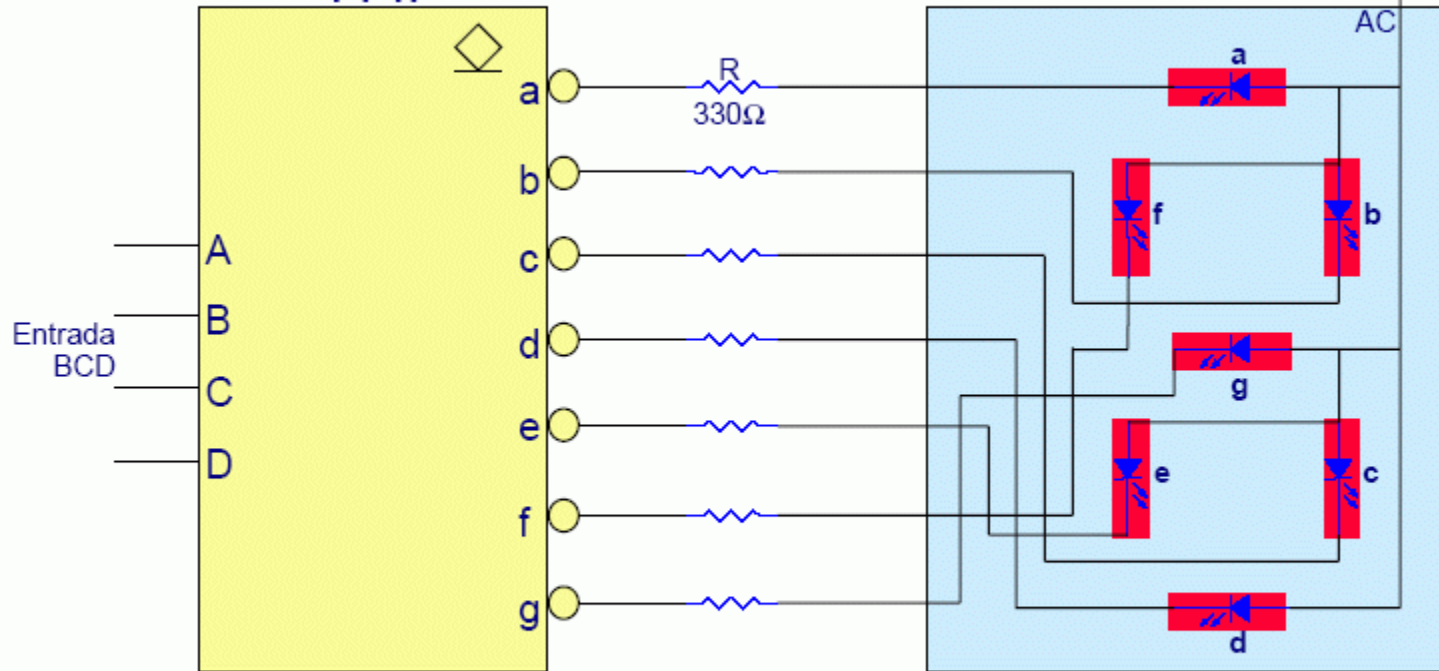
LOGIC SYMBOL



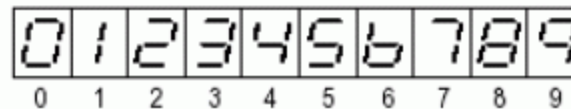
V_{CC} = PIN 16
GND = PIN 8

Salidas en colector abierto

74'47



Visualizador de 7 segmentos



RBI	0.5 U.L.	0.25 U.L.
LT	0.5 U.L.	0.25 U.L.
BI/RBO	0.5 U.L.	0.75 U.L.
a, to g	1.2 U.L.	2.0 U.L.
	Open-Collector	15 (7.5) U.L.

NOTES:

a) 1 Unit Load (U.L.) = 40 μ A HIGH, 1.6 mA LOW.

b) Output current measured at V_{OUT} = 0.5 V

The Output LOW drive factor is 7.5 U.L. for Military (54) and 15 U.L. for Commercial (74) Temperature Ranges.

BCD TO 7-SEGMENT DECODER

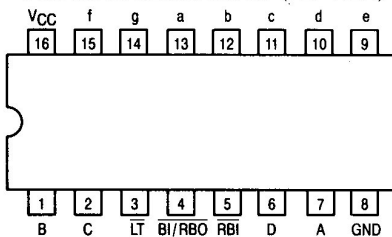
The SN54/74LS48 is a BCD to 7-Segment Decoder consisting of NAND gates, input buffers and seven AND-OR-INVERT gates. Seven NAND gates and one driver are connected in pairs to make BCD data and its complement available to the seven decoding AND-OR-INVERT gates. The remaining NAND gate and three input buffers provide lamp test, blanking input/ripple-blanking input for the LS48.

The circuit accepts 4-bit binary-coded-decimal (BCD) and, depending on the state of the auxiliary inputs, decodes this data to drive other components. The relative positive logic output levels, as well as conditions required at the auxiliary inputs, are shown in the truth tables.

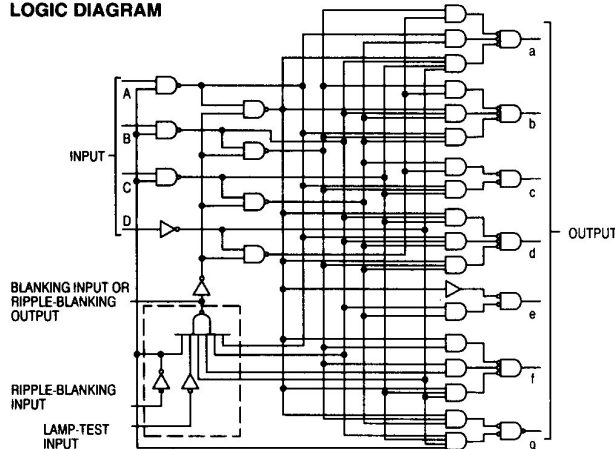
The LS48 circuit incorporates automatic leading and/or trailing edge zero-blanking control (RBI and RBO). Lamp Test (LT) may be activated any time when the BI/RBO node is HIGH. Both devices contain an overriding blanking input (BI) which can be used to control the lamp intensity by varying the frequency and duty cycle of the BI input signal or to inhibit the outputs.

- Lamp Intensity Modulation Capability (BI/RBO)
- Internal Pull-Ups Eliminate Need for External Resistors
- Input Clamp Diodes Eliminate High-Speed Termination Effects

CONNECTION DIAGRAM DIP (TOP VIEW)



LOGIC DIAGRAM

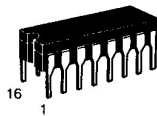


SN54/74LS48

Cátodo comum

BCD TO 7-SEGMENT DECODER

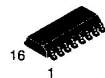
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

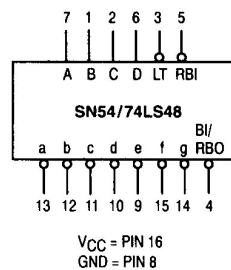


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

SN54LSXXJ Ceramic
SN74LSXXN Plastic
SN74LSXXD SOIC

LOGIC SYMBOL



SEGMENT /DRIVER

are Low Power Schottky BCD to 7-Segment Decoder NAND gates, input buffers and seven AND-OR-INVERT gates. The remaining NAND gate and three input buffers provide lamp test, blanking input/ripple-blanking output and ripple-blanking output.

-bit binary-coded-decimal (BCD) and, depending on inputs, decodes this data to drive a 7-segment display positive-logic output levels, as well as conditions try inputs, are shown in the truth tables. Output V54/74LS47 are designed to withstand the relatively for 7-segment indicators.

withstand 15 V with a maximum reverse current of outputs requiring up to 24 mA of current may be driven LS47 high performance output transistors. Display counts above nine are unique symbols to authenticate

ncorporates automatic leading and/or trailing-edge (BI and RBO). Lamp test (LT) may be performed at any mode is a HIGH level. This device also contains an t (BI) which can be used to control the lamp intensity y and duty cycle of the BI input signal or to inhibit the

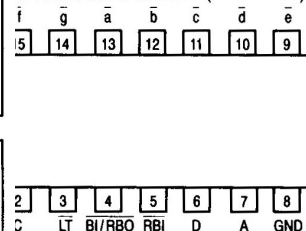
ilation Capability (BI/RBO)

uts

o Suppression

Limit High-Speed Termination Effects

CONNECTION DIAGRAM DIP (TOP VIEW)



LOADING (Note a)

	HIGH	LOW
Inputs	0.5 U.L.	0.25 U.L.
-Blanking Input	0.5 U.L.	0.25 U.L.
Test Input	0.5 U.L.	0.25 U.L.
ng Input or	0.5 U.L.	0.75 U.L.
-Blanking Output	1.2 U.L.	2.0 U.L.
ts	Open-Collector	15 (7.5) U.L.

HIGH, 1.6 mA LOW.

t VOUT = 0.5 V

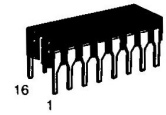
or is 7.5 U.L. for Military (54) and 15 U.L. for Commercial (74) Temperature Ranges.

SN54/74LS47

Ânodo comum

BCD TO 7-SEGMENT DECODER/DRIVER

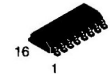
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

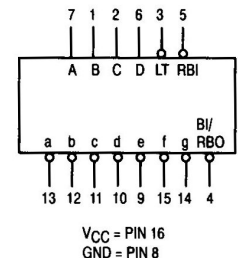


D SUFFIX
SOIC
CASE 751B-03

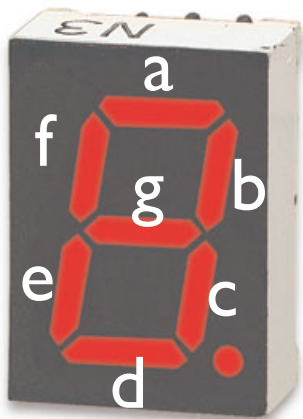
ORDERING INFORMATION

SN54LSXXJ Ceramic
SN74LSXXN Plastic
SN74LSXXD SOIC

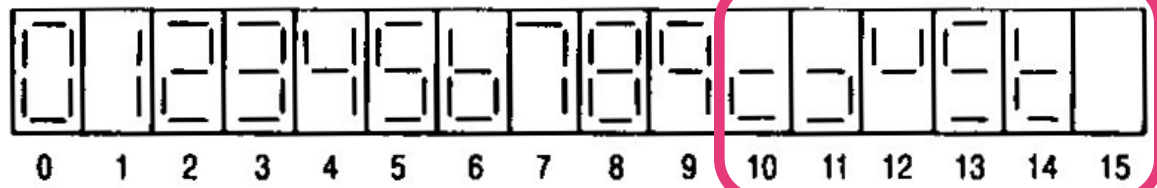
LOGIC SYMBOL



DEC Display 7 Segmentos



Caracteres formados para
códigos **BCD** “inválidos”



NUMERICAL DESIGNATIONS — RESULTANT DISPLAYS



BCD TO 7-SEGMENT DECODER

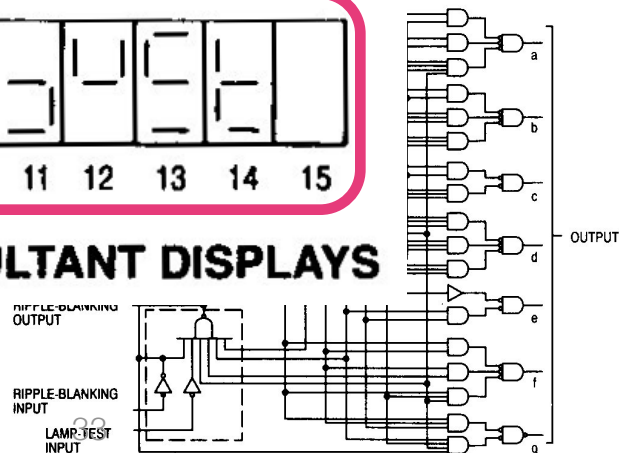
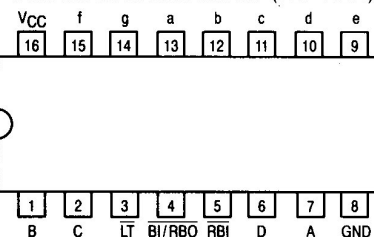
The SN54/74LS48 is a BCD to 7-Segment Decoder consisting of NAND gates, input buffers and seven AND-OR-INVERT gates. Seven NAND gates and one driver are connected in pairs to make BCD data and its complement available to the seven decoding AND-OR-INVERT gates. The remaining NAND gate and three input buffers provide lamp test, blanking input/ripple-blanking input for the LS48.

The circuit accepts 4-bit binary-coded-decimal (BCD) and, depending on the state of the auxiliary inputs, decodes this data to drive other components. The relative positive logic output levels, as well as conditions required at the auxiliary inputs, are shown in the truth tables.

The LS48 circuit incorporates automatic leading and/or trailing edge zero-blanking control (RBI and RBO). Lamp Test (LT) may be activated any time when the BI/RBO node is HIGH. Both devices contain an overriding blanking input (BI) which can be used to control the lamp intensity by varying the frequency and duty cycle of the BI input signal or to inhibit the outputs.

- Lamp Intensity Modulation Capability (BI/RBO)
- Internal Pull-Ups Eliminate Need for External Resistors
- Input Clamp Diodes Eliminate High-Speed Termination Effects

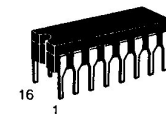
CONNECTION DIAGRAM DIP (TOP VIEW)



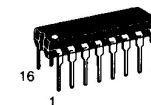
SN54/74LS48

BCD TO 7-SEGMENT DECODER

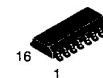
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

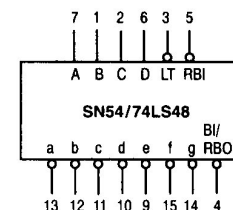


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

SN54LSXXJ	Ceramic
SN74LSXXN	Plastic
SN74LSXXD	SOIC

LOGIC SYMBOL

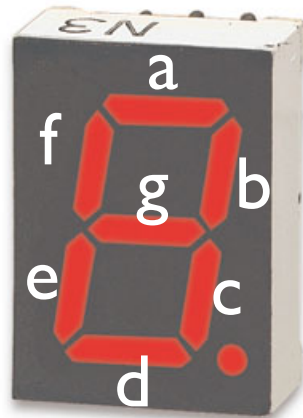


VCC = PIN 16
GND = PIN 8

Condições “Don't care”

- Alguns circuitos possuem certas combinações de entrada para as quais não importa a saída do circuito (ou não existe especificação do nível lógico de saída).
- Ou seja, certas combinações de entradas que não geram impacto na função lógica (a saída pode ir tanto para nível lógico ALTO quanto BAIXO).
- Recebem o nome de termos “don't care”. Na tabela verdade e no mapa, estas condições são registradas usando a letra “X”.
- Exemplo: (segue)...

DEC Display 7 Segmentos/BCD



DECIMAL	BCD
	8421
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

Digit Shown	Illuminated Segment (1 = illumination)						
	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	0	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	1	0	1	1

Decimal	Entradas				Saídas						
	D	C	B	A	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1		1	1				
2	0	0	1	0	1	1		1	1		1
3	0	0	1	1	1	1	1	1			1
4	0	1	0	0		1	1			1	1
5	0	1	0	1	1		1	1		1	1
6	0	1	1	0	1		1	1	1	1	1
7	0	1	1	1	1	1	1				
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1			1	1
10	1	0	1	0	X	X	X	X	X	X	X
11	1	0	1	1	X	X	X	X	X	X	X
12	1	1	0	0	X	X	X	X	X	X	X
13	1	1	0	1	X	X	X	X	X	X	X
14	1	1	1	0	X	X	X	X	X	X	X
15	1	1	1	1	X	X	X	X	X	X	X

Condições “don't care”

DEC Display 7 Segmentos/BCD

Decimal	Entradas				Saídas						
	D	C	B	A	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1		1	1				
2	0	0	1	0	1	1		1	1		1
3	0	0	1	1	1	1	1	1			1
4	0	1	0	0		1	1			1	1
5	0	1	0	1	1		1	1		1	1
6	0	1	1	0	1		1	1	1	1	1
7	0	1	1	1	1	1	1				
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1			1	1
10	1	0	1	0	X	X	X	X	X	X	X
11	1	0	1	1	X	X	X	X	X	X	X
12	1	1	0	0	X	X	X	X	X	X	X
13	1	1	0	1	X	X	X	X	X	X	X
14	1	1	1	0	X	X	X	X	X	X	X
15	1	1	1	1	X	X	X	X	X	X	X

Saída = “X” : significa
indiferente

Condições indiferentes

Projeto DEC Display 7 Segmentos

D	C	B	A	a	b	c	d	e	f	g	Display
0	0	0	0	1	1	1	1	1	1	0	"0"
0	0	0	1	0	1	1	0	0	0	0	"1"
0	0	1	0	1	1	0	1	1	0	1	"2"
0	0	1	1	1	1	1	1	0	0	1	"3"
0	1	0	0	0	1	1	0	0	1	1	"4"
0	1	0	1	1	0	1	1	0	1	1	"5"
0	1	1	0	1	0	1	1	1	1	1	"6"
0	1	1	1	1	1	1	0	0	0	0	"7"
1	0	0	0	1	1	1	1	1	1	1	"8"
1	0	0	1	1	1	1	1	0	1	1	"9"

D	C	B	A	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1

Sem "don't care":

a:

BA \ DC	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11				
10	1	1		

Com "don't care":

a:

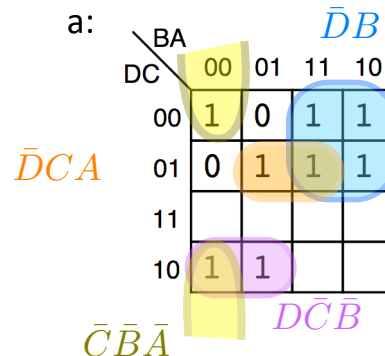
BA \ DC	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

Projeto DEC Display 7 Segmentos

D	C	B	A	a	b	c	d	e	f	g	Display
0	0	0	0	1	1	1	1	1	1	0	"0"
0	0	0	1	0	1	1	0	0	0	0	"1"
0	0	1	0	1	1	0	1	1	0	1	"2"
0	0	1	1	1	1	1	1	0	0	1	"3"
0	1	0	0	0	1	1	0	0	1	1	"4"
0	1	0	1	1	0	1	1	0	1	1	"5"
0	1	1	0	1	0	1	1	1	1	1	"6"
0	1	1	1	1	1	1	0	0	0	0	"7"
1	0	0	0	1	1	1	1	1	1	1	"8"
1	0	0	1	1	1	1	1	0	1	1	"9"

D	C	B	A	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1

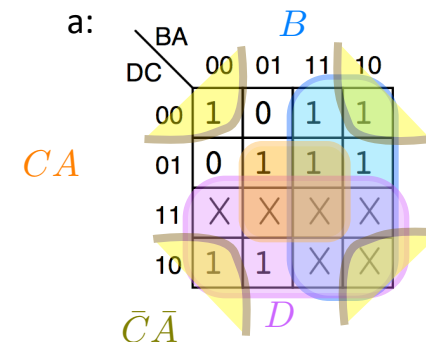
Sem "don't care":



$$a = \bar{D}B + \bar{D}CA + \bar{C}\bar{B}\bar{A} + D\bar{C}\bar{B}$$

1xOR(4)+1xAND(2)+3xAND(3)+4xNOT

Com "don't care":



$$a = B + CA + \bar{C}\bar{A} + D$$

1xOR(4)+2xAND(2)+2xNOT

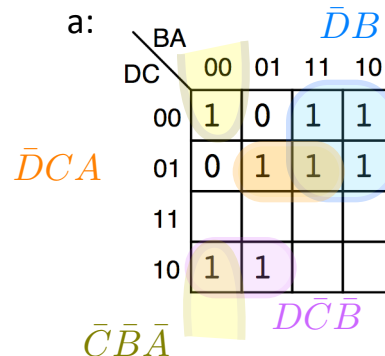
Projeto DEC Display 7 Segmentos

D	C	B	A	a	b	c	d	e	f	g	Display
0	0	0	0	1	1	1	1	1	1	0	"0"
0	0	0	1	0	1	1	0	0	0	0	"1"
0	0	1	0	1	1	0	1	1	0	1	"2"
0	0	1	1	1	1	1	1	0	0	1	"3"
0	1	0	0	0	1	1	0	0	1	1	"4"
0	1	0	1	1	0	1	1	0	1	1	"5"
0	1	1	0	1	0	1	1	1	1	1	"6"
0	1	1	1	1	1	1	0	0	0	0	"7"
1	0	0	0	1	1	1	1	1	1	1	"8"
1	0	0	1	1	1	1	1	0	1	1	"9"

Desvantagem:

- Segmento "a" acende também para códigos não-BCD (ou códigos BCD inválidos)!
- Juntar este "sintoma" ao que acontecerá com os outros segmentos => formação de caracteres estranhos para códigos BCD inválidos!

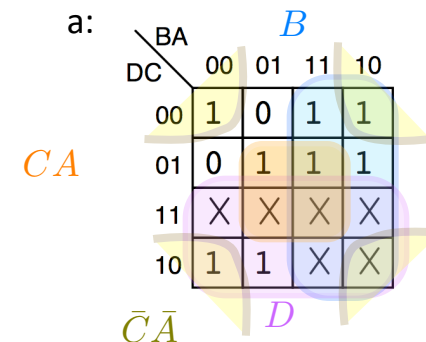
Sem "don't care":



$$a = \bar{D}B + \bar{D}CA + \bar{C}\bar{B}\bar{A} + D\bar{C}\bar{B}$$

1xOR(4)+1xAND(2)+3xAND(3)+4xNOT

Com "don't care":



$$a = B + CA + \bar{C}\bar{A} + D$$

1xOR(4)+2xAND(2)+2xNOT

Projeto DEC Display 7 Segmentos

D	C	B	A	a	b	c	d	e	f	g	Display
0	0	0	0	1	1	1	1	1	1	0	"0"
0	0	0	1	0	1	1	0	0	0	0	"1"
0	0	1	0	1	1	0	1	1	0	1	"2"
0	0	1	1	1	1	1	1	0	0	1	"3"
0	1	0	0	0	1	1	0	0	1	1	"4"
0	1	0	1	1	0	1	1	0	1	1	"5"
0	1	1	0	1	0	1	1	1	1	1	"6"
0	1	1	1	1	1	1	0	0	0	0	"7"
1	0	0	0	1	1	1	1	1	1	1	"8"
1	0	0	1	1	1	1	1	0	1	1	"9"

Com "don't care":

AB\CD	00	01	11	10
00				
01				
11				
10				

$$e = \bar{C} \cdot \bar{A} + B \cdot \bar{A}$$

AB\CD	00	01	11	10
00				
01				
11				
10				

$$g = D + \underbrace{\bar{C} \cdot B + C \cdot \bar{B}}_{C \oplus B} + C \cdot \bar{A}$$

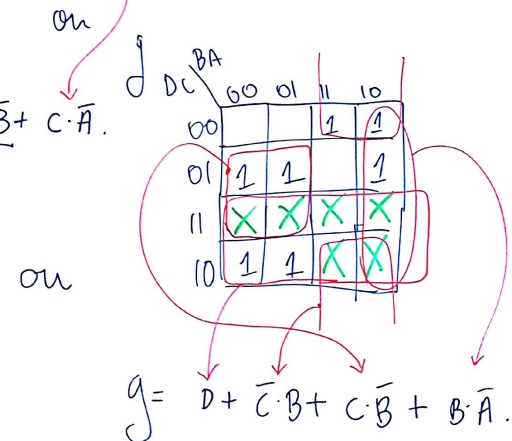
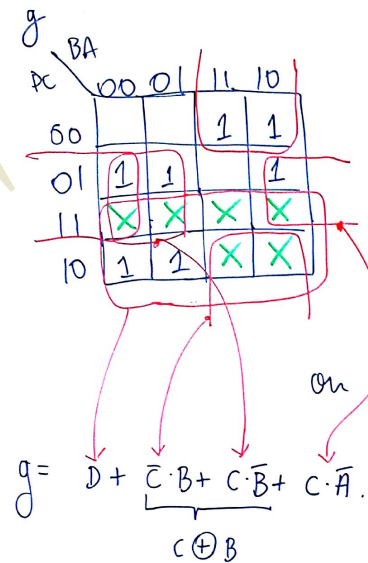
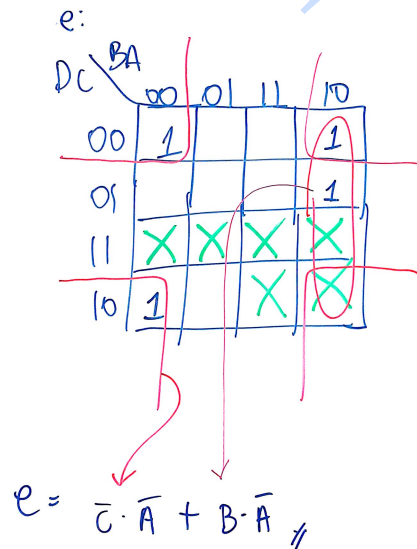
ou

$$g = D + \bar{C} \cdot B + C \cdot \bar{B} + B \cdot \bar{A}$$

Projeto DEC Display 7 Segmentos

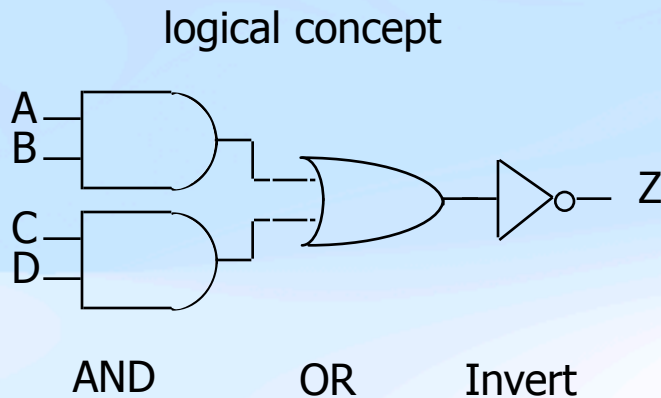
D	C	B	A	a	b	c	d	e	f	g	Display
0	0	0	0	1	1	1	1	1	1	0	"0"
0	0	0	1	0	1	1	0	0	0	0	"1"
0	0	1	0	1	1	0	1	1	0	1	"2"
0	0	1	1	1	1	1	1	0	0	1	"3"
0	1	0	0	0	1	1	0	0	1	1	"4"
0	1	0	1	1	0	1	1	0	1	1	"5"
0	1	1	0	1	0	1	1	1	1	1	"6"
0	1	1	1	1	1	1	0	0	0	0	"7"
1	0	0	0	1	1	1	1	1	1	1	"8"
1	0	0	1	1	1	1	1	0	1	1	"9"

Com “don’t care”:

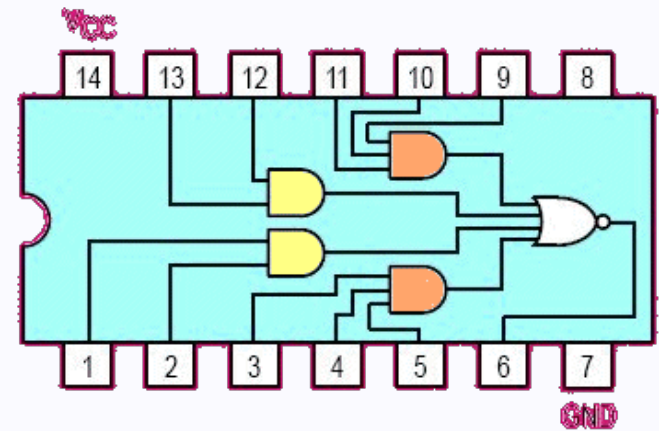
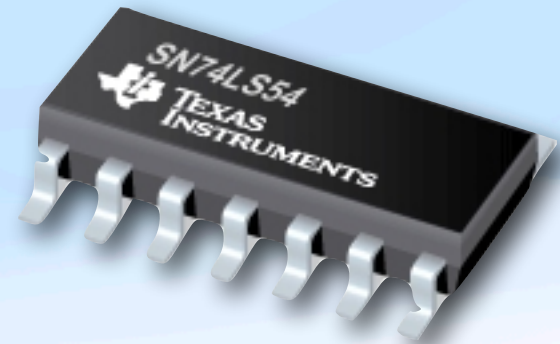
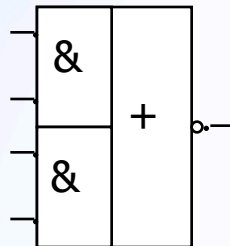


USO de AOIs para Implementar Funções

■ AOI = And + Or + Inverter (NOT):



2x2 AOI gate
symbol



74LS54

USO de AOIs para Implementar Funções

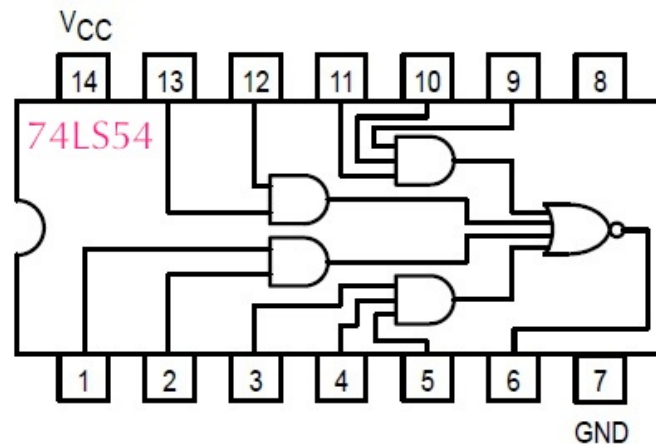
■ Exemplo: $a = B + CA + \bar{C}\bar{A} + D$

A •

B •

C •

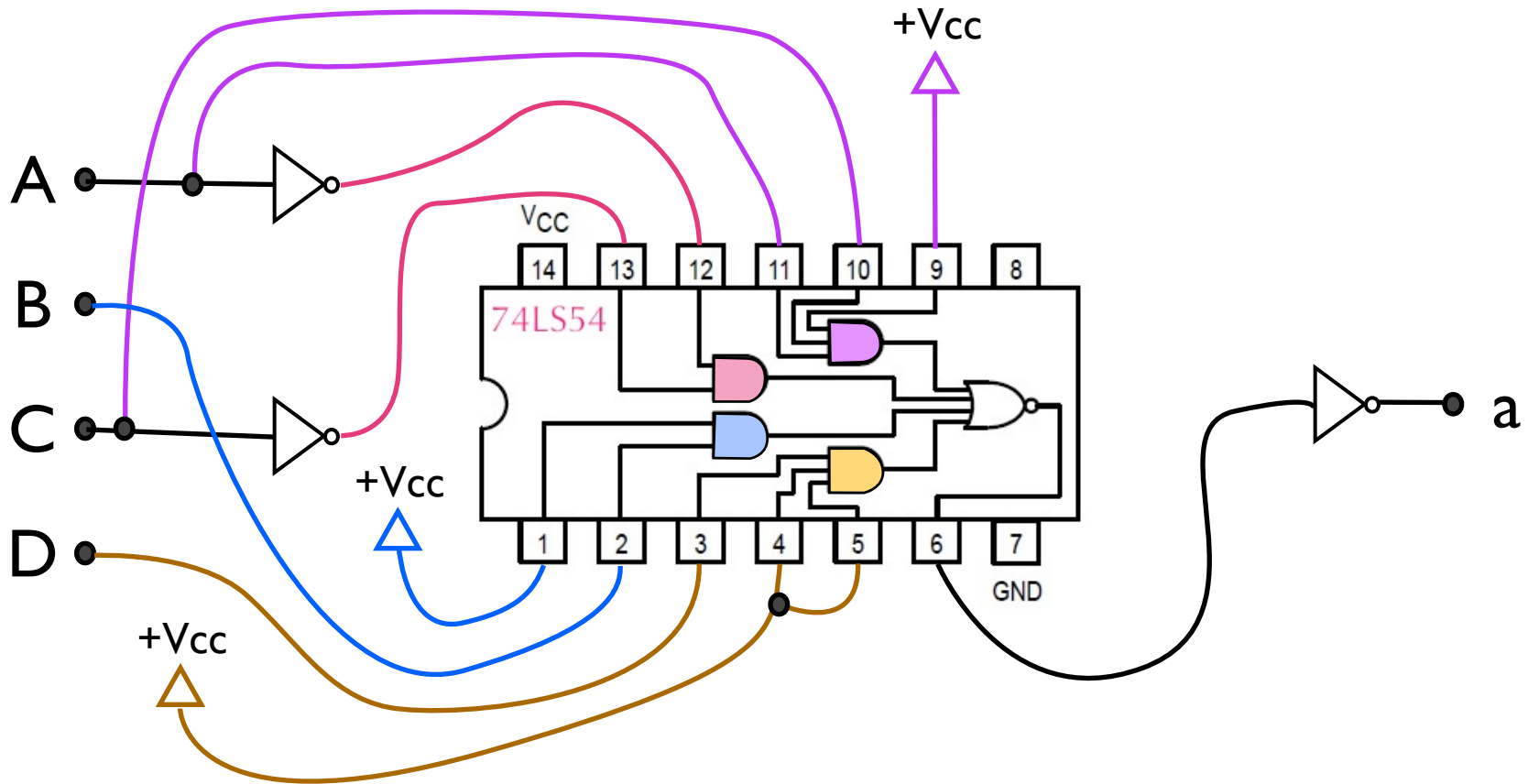
D •



• a

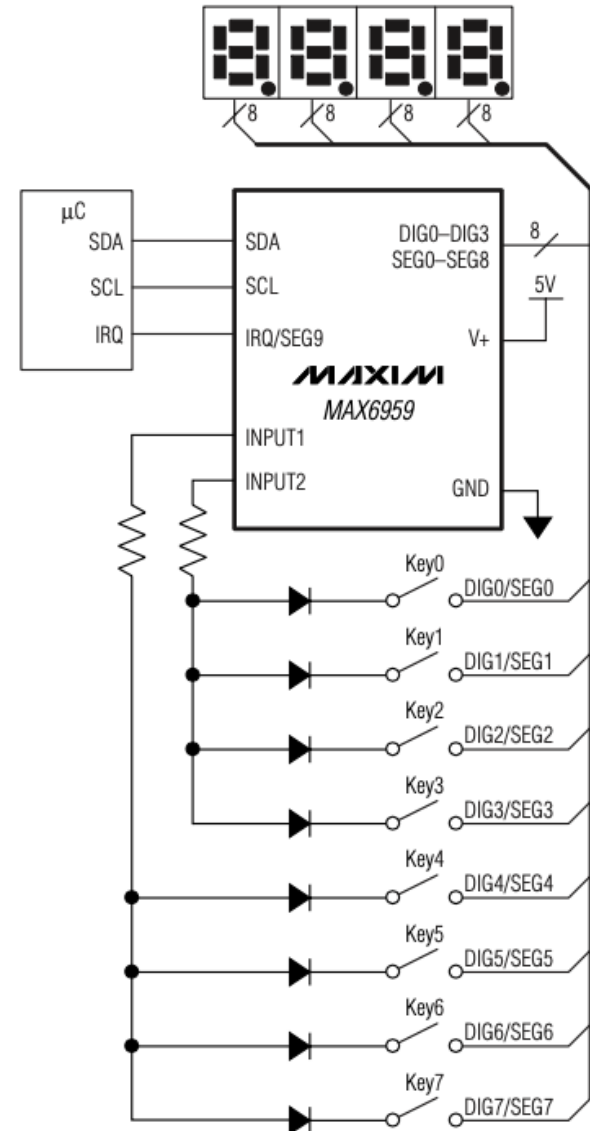
USO de AOs para Implementar Funções

■ Exemplo: $a = B + CA + \bar{C}\bar{A} + D$



Outros DEC Display/Hexa...

- EDE707
- MAX6958/6959:
Display cátodo comum, 7 ou 9 segmentos;
Multiplexa até 4 displays;
I²C (2-wire serial interface);
Compatível com sistemas de 2,5V e 3,3V
- Hexadecimal-to-Seven Segment Driver:
MC14495 (CMOS; Motorola; 16 Pin DIP)
- DM9368: 7-Segment Decoder/Driver/Latch
with Constant Current Source Outputs
- 4311 ou 4368 (CMOS)



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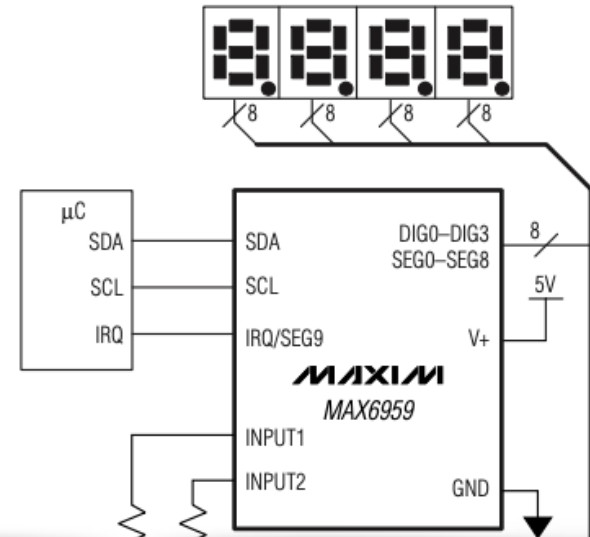


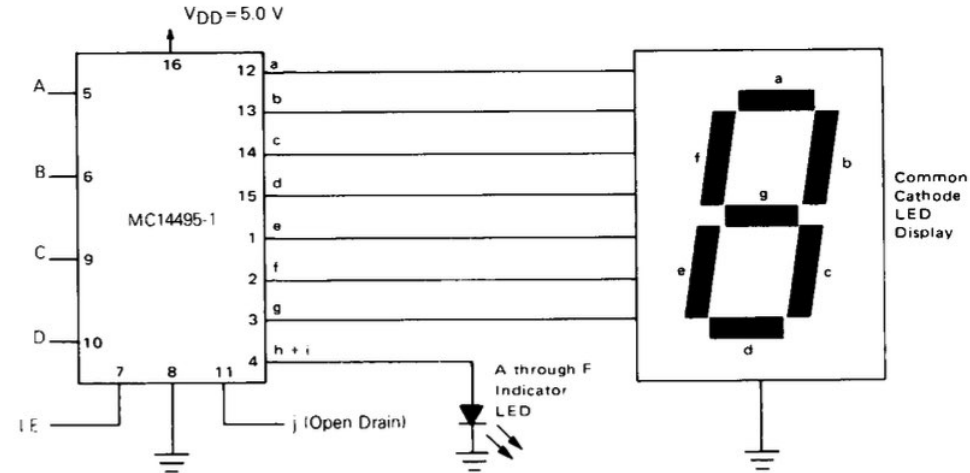
Table 8. Seven-Segment Mapping Decoder for Hexadecimal Font

7-SEGMENT CHARACTER	REGISTER DATA					ON SEGMENTS = 1						
	D7-D4	D3	D2	D1	D0	a	b	c	d	e	f	g
0	X	0	0	0	0	1	1	1	1	1	1	0
1	X	0	0	0	1	0	1	1	0	0	0	0
2	X	0	0	1	0	1	1	0	1	1	0	1
3	X	0	0	1	1	1	1	1	1	0	0	1
4	X	0	1	0	0	0	1	1	0	0	1	1
5	X	0	1	0	1	1	0	1	1	0	1	1
6	X	0	1	1	0	1	0	1	1	1	1	1
7	X	0	1	1	1	1	1	1	0	0	0	0
8	X	1	0	0	0	1	1	1	1	1	1	1
9	X	1	0	0	1	1	1	1	1	0	1	1
A	X	1	0	1	0	1	1	1	0	1	1	1
B	X	1	0	1	1	0	0	1	1	1	1	1
C	X	1	1	0	0	1	0	0	1	1	1	0
D	X	1	1	0	1	0	1	1	1	1	0	1
E	X	1	1	1	0	1	0	0	1	1	1	1
F	X	1	1	1	1	1	0	0	0	1	1	1

MAX6958/MAX6959

Outros DEC Display/Hexa...

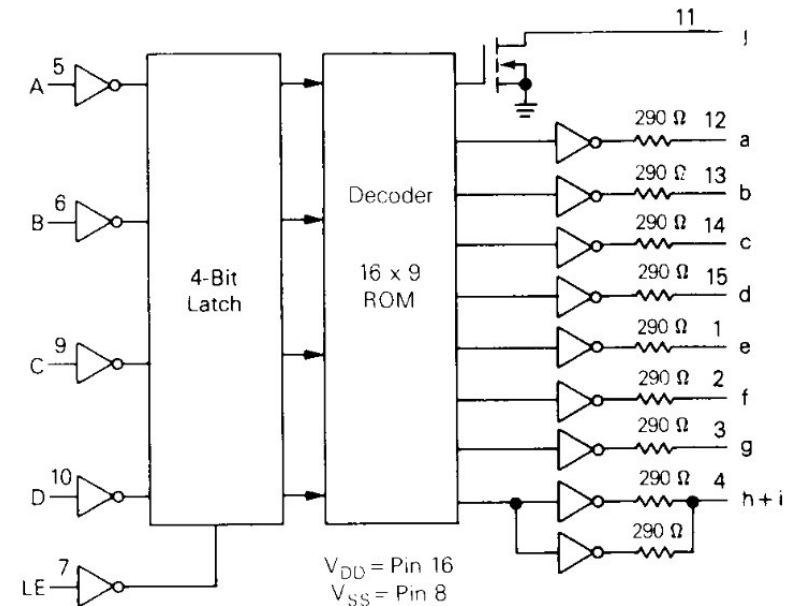
- EDE707
- MAX6958/6959:
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with Constant Current Source Outputs
- 4311 ou 4368 (CMOS)



ALPHANUMERIC DISPLAY

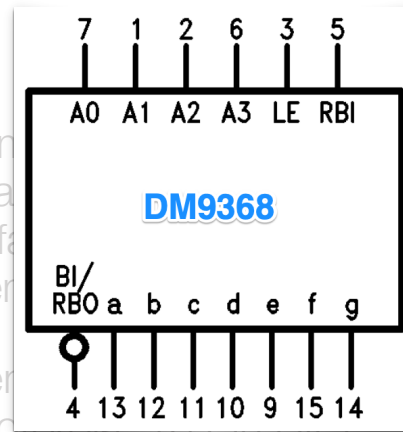
0 1 2 3 4 5 6 7 8 9 A B C d e F

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15



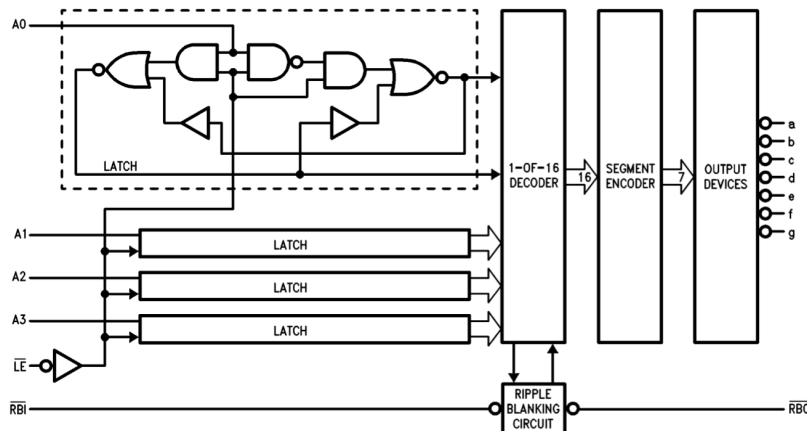
Outros DEC Display/Hexa...

- EDE707
- MAX6958/6959:
Display cátodo comum
Multiplexa até 4 displays
I²C (2-wire serial interface)
Compatível com sistemas de 5V e 3.3V
- Hexadecimal-to-Seven-Segment Decoder/Driver
MC14495 (CMOS; Motorola; 16 Pins DIP)
- DM9368: 7-Segment Decoder/Driver/Latch with Constant Current Source Outputs



BINARY STATE	INPUTS						OUTPUTS								DISPLAY
	$\overline{LE}$	$\overline{RBI}$	A3	A2	A1	A0	a	b	c	d	e	f	g	$\overline{RBO}$	
—	H	*	X	X	X	X	← STABLE →							H	STABLE
0	L	L	L	L	L	L	L	L	L	L	L	L	L	L	BLANK
0	L	H	L	L	L	L	H	H	H	H	H	H	L	H	0
1	L	X	L	L	L	H	L	H	H	L	L	L	L	H	1
2	L	X	L	L	H	L	H	H	L	H	H	L	H	H	2
3	L	X	L	L	H	H	H	H	H	L	L	L	H	H	3
4	L	X	L	H	L	L	L	H	H	L	L	H	H	H	4
5	L	X	L	H	L	H	H	L	H	H	L	H	H	H	5
6	L	X	L	H	H	L	H	L	H	H	H	H	H	H	6
7	L	X	L	H	H	H	H	H	H	L	L	L	L	H	7
8	L	X	H	L	L	L	H	H	H	H	H	H	H	H	8
9	L	X	H	L	L	H	H	H	H	L	L	H	H	H	9
10	L	X	H	L	H	L	H	H	H	L	H	H	H	H	A
11	L	X	H	L	H	H	L	L	H	H	H	H	H	H	b
12	L	X	H	H	L	L	H	L	L	H	H	H	L	H	c
13	L	X	H	H	L	H	L	H	H	H	H	L	H	H	d
14	L	X	H	H	H	L	H	L	L	H	H	H	H	H	e
15	L	X	H	H	H	H	H	L	L	L	H	H	H	H	f
X	X	X	X	X	X	X	L	L	L	L	L	L	L	L**	BLANK

Logic Diagram



- *The $\overline{RBI}$ will blank the display only if a binary zero is stored in the latches.
- **The $\overline{RBO}$ used as an input overrides all other input conditions.

Pin Name	Description
A0–A3	Address (Data) Inputs
$\overline{RBO}$	Ripple Blanking Output (Active Low)
$\overline{RBI}$	Ripple Blanking Input (Active Low)
a–g	Segment Drivers-Outputs
$\overline{LE}$	Latch Enable Input (Active Low)

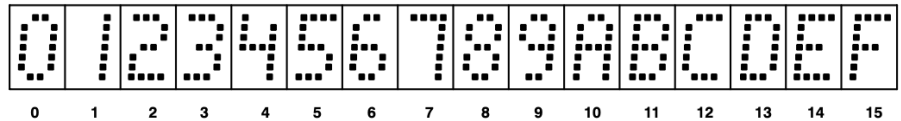
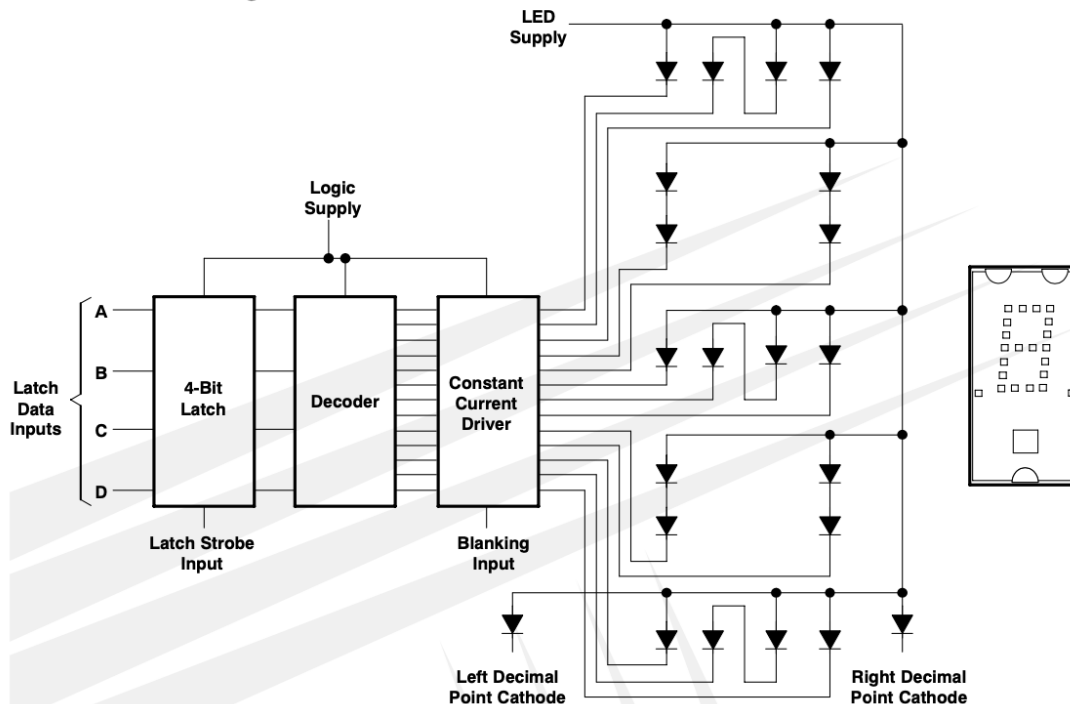
TL/F/9796–8

Display HEXA c/DEC

TIL311

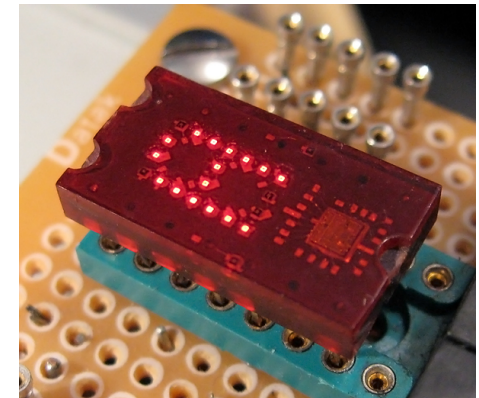
- Internal TTL MSI IC with Latch, Decoder, and Driver
- 0.300-Inch (7,62-mm) Character Height
- Wide Viewing Angle
- High Brightness
- Left-and-Right-Hand Decimals
- Constant-Current Drive for Hexadecimal Characters
- Separate LED and Logic Power Supplies May Be Used
- Operates from 5-V Supply

Functional Block Diagram



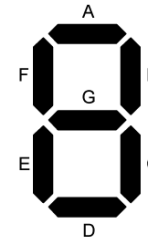
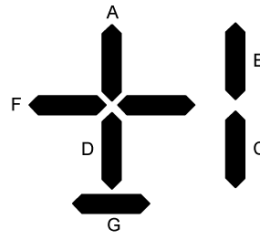
PACKAGE P 14-LEAD PDIP (BOTTOM VIEW)

Logic Supply	14	1 LED Supply
Data Input C	13	2 Data Input B
Data Input D	12	3 Data Input A
omitted	11	4 Left Decimal
Right Decimal	10	5 Strobe Input
omitted	9	6 omitted
Blanking Input	8	7 Ground

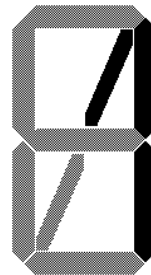


Outros Displays

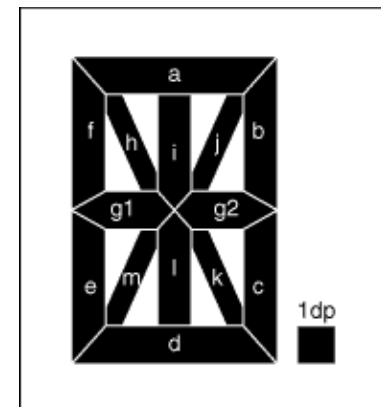
■ 1/2 Display:



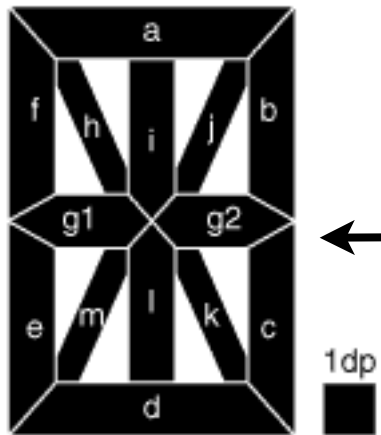
■ 9 segmentos:



■ 14/16 segmentos.



Outros Displays



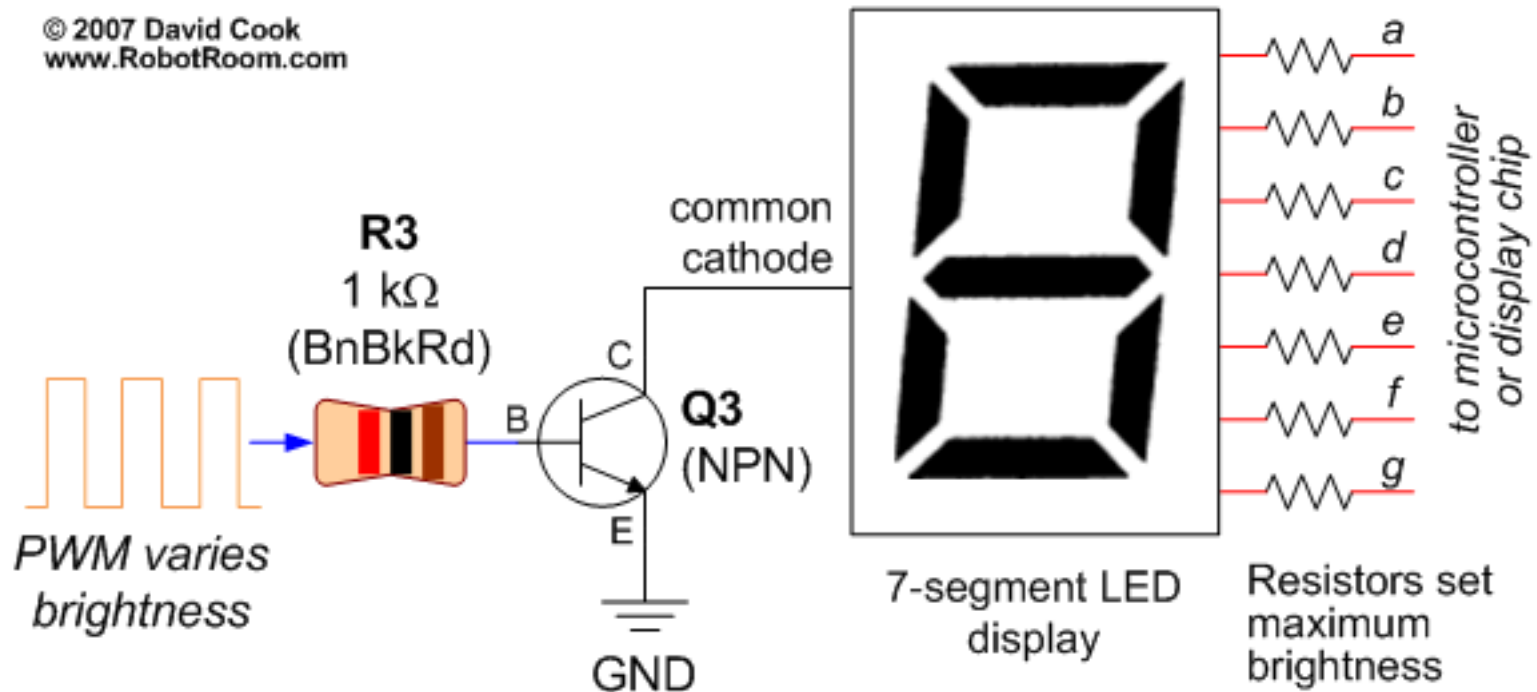
← 14/16 segmentos →



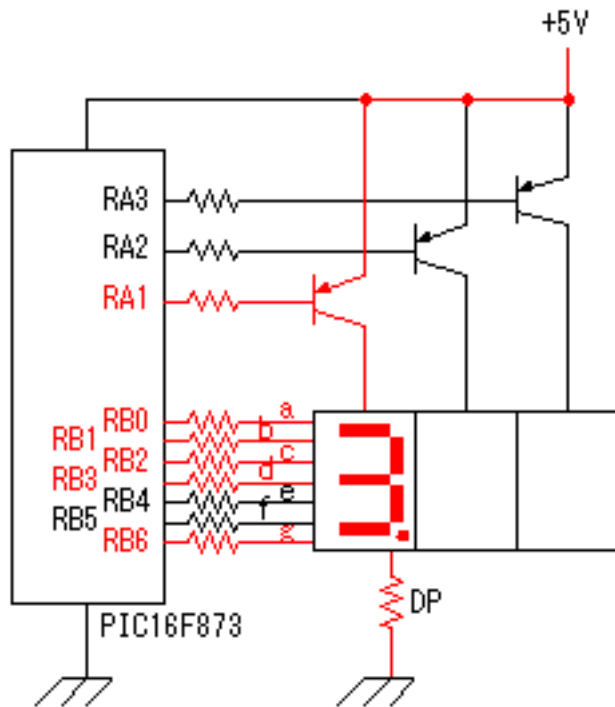
MSB LSB	x000	x001	x010	x011	x100	x101	x110	x111
0000								
0001								
0010								
0011								
0100								
0101								
0110								
0111								
1000								
1001								
1010								
1011								
1100								
1101								
1110								
1111								

Controlando Brilho Display

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www.RobotRoom.com

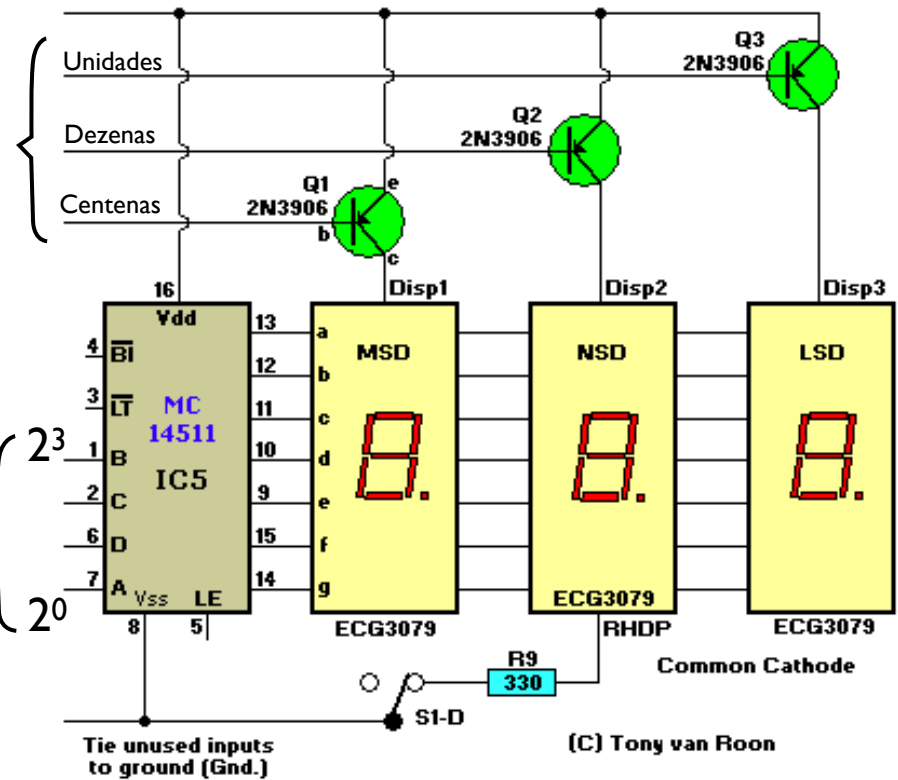


Multiplexação de Displays



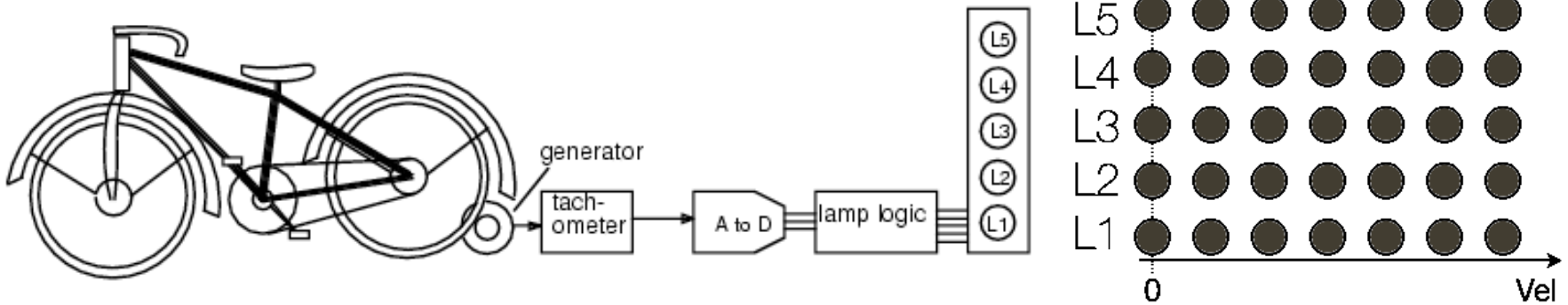
Linhas habilitação de cada display

Código BCD



Problema:

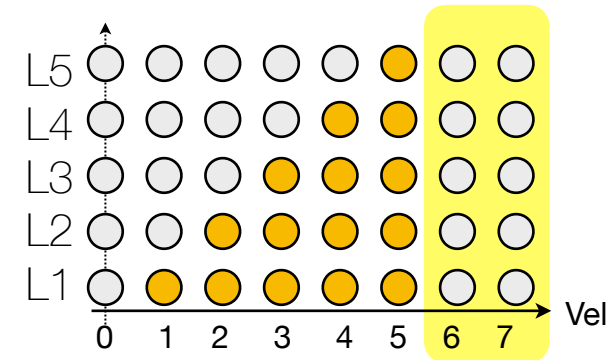
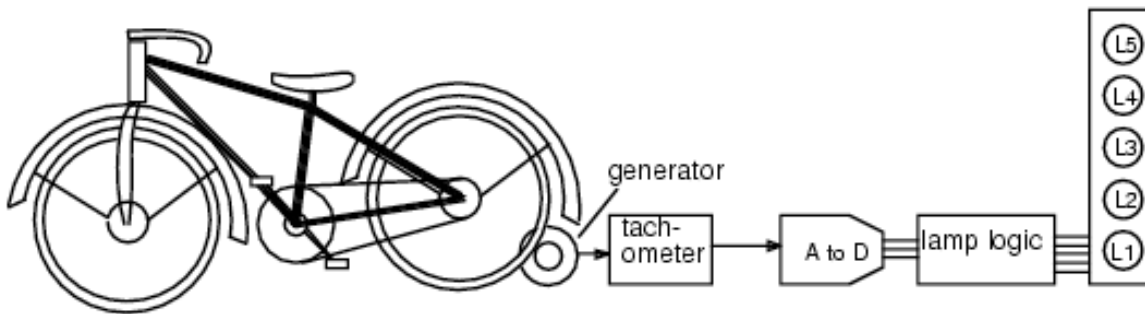
- Projetar um circuito onde o número de leds ativos aumenta conforme aumenta a velocidade da bicicleta.



- Neste circuito um pequeno gerador DC acoplado à roda da bicicleta gera uma saída proporcional a velocidade sendo desenvolvida.
- A saída do gerador DC alimenta a placa de um tacogerador que limita sua tensão de saída se o gerador DC ultrapassar certa tensão. Isto é, mesmo que a velocidade da bicicleta aumente além de certa velocidade máxima, a saída do tacogerador já atingiu seu limite e todos os leds já estarão acesos.
- Um conversor A/D de 3-bits converte a tensão do tacogerador para os bits de saída: A, B e C, onde A=MSB e C=LSB.
- Detalhes: o conjunto de 5 leds deve responder para 6 códigos de saída do A/D. Quando ABC=000, bicicleta sem movimento, nenhum led deve se ativar. Para os seguintes 5 códigos de saída: 001 ~ 101 os leds L1, L2, L3, L4 e L5 devem se ativar conforme a velocidade aumenta. Note que não estamos usando os códigos de saída: 110 e 111 porque se supõe que estes códigos nunca ocorrem uma vez que a faixa de excursão de entrada do A/D foi limitada pela placa do tacogerador.

Problema:

- Projetar um circuito onde o número de leds ativos aumenta conforme aumenta a velocidade da bicicleta.

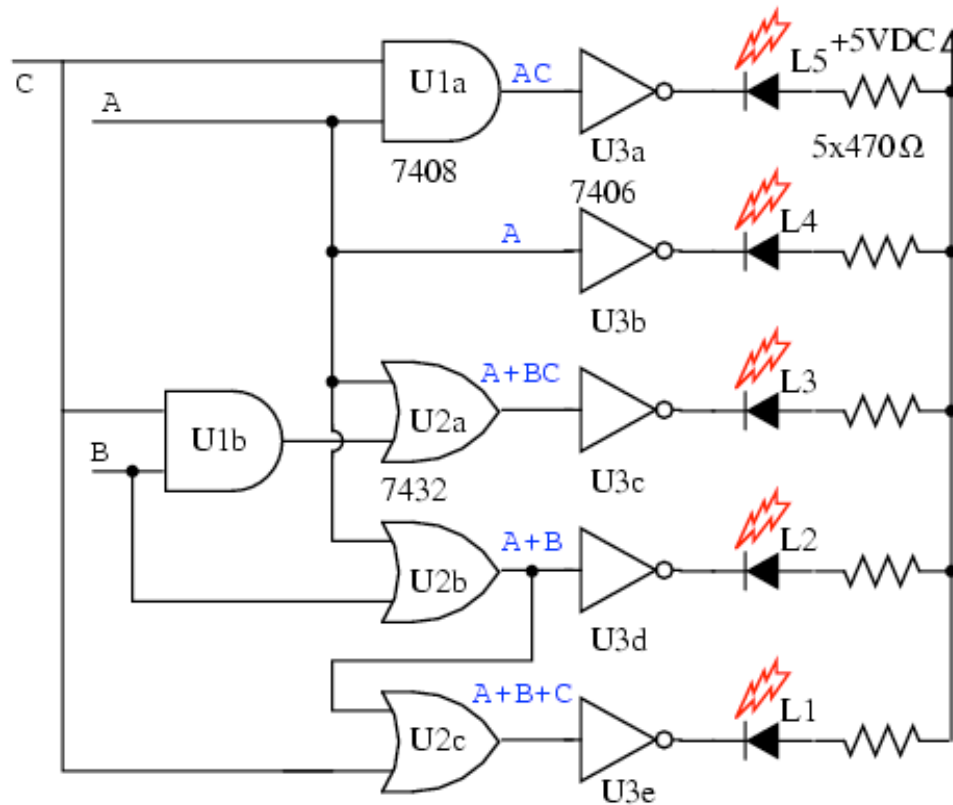
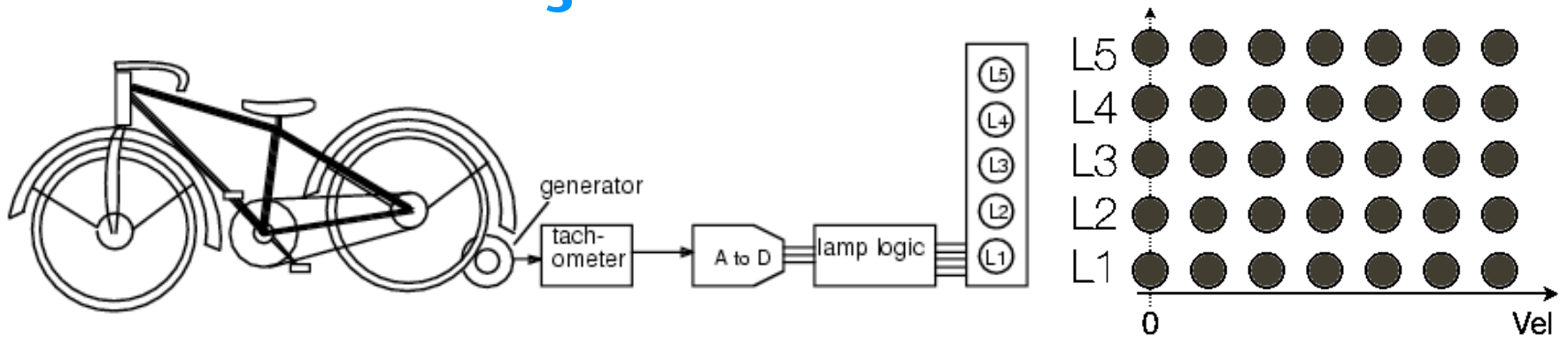


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Tabela Verdade:

Ref	ABC	L1	L2	L3	L4	L5
0	000	0	0	0	0	0
1	001	1	0	0	0	0
2	010	1	1	0	0	0
3	011	1	1	1	0	0
4	100	1	1	1	1	0
5	101	1	1	1	1	1
6	110	X	X	X	X	X
7	111	X	X	X	X	X

Solução Possível:





You have to fall,
before you can fly